Control of Electro-Mechanical Systems
Lecture notes

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Chapter 1

Dynamic systems and the frequency domain

In a system, we distinguish the following quantities:

- the state x;
- the input u, which is the set of variables that we can directly control;
- the output y, which is a set of variables that we measure and/or for which we want to achieve some particular control objective;
- possibly, we also introduce a disturbance d, which is an additional input that we cannot control.

1.1 Time domain representation

In the continuous time domain, the evolution of a system is governed by a differential equation, and the output is a function of the state and input

$$\begin{aligned} \dot{x} &= f(x) + g(x, u) \\ y &= h(x, u) \end{aligned} \tag{1.1}$$

If the system is linear, it is usually notated as

$$\begin{aligned} \dot{x} &= Ax + Bu \\ y &= Cx + Du \end{aligned} \tag{1.2}$$

1.2 Laplace transforms and frequency domain

A different way of representing a system, is the frequency-domain representation. For this, we make use of the Laplace transform. Given a signal $f(t)$ expressed

as a function of time, its Laplace transform is

$$F(s) = \int_0^\infty e^{-st} f(t) dt \tag{1.3}$$

where s is a complex number. Laplace transforms of signals are usually notated using uppercase letters corresponding to the lowercase letter of the time-domain signal.

1.3 Properties of the Laplace transform

This section will describe the most important properties of the Laplace transform:

- Linearity
- Translation in time
- Derivation
- Integration
- Convolution
- Final value theorem

1.3.1 Linearity

The property of *linearity* states that the Laplace transform of a linear composition of functions is equal to the composition of the Laplace transforms of the individual functions.

$$\mathcal{L}\left[a \cdot f(t) + b \cdot g(t)\right] = a \cdot \mathcal{L}\left[f(t)\right] + b \cdot \mathcal{L}\left[g(t)\right] \tag{1.4}$$

This can be seen immediately by applying the definition:

$$\int_0^\infty \left(af(t) + bg(t)\right) e^{-st} dt = a \int_0^\infty f(t) e^{-st} dt + b \int_0^\infty g(t) e^{-st} dt \tag{1.5}$$

1.3.2 Time shift

The property of *time shifting* is concerned with what happens if we take a function $f(t)$ and apply a translation in time T. This property only applies if the signal is such that $f(t) = 0$ for $t < 0$. The transform of the shifted function $f(t-T)$ can be expressed as the transform of the unshifted function multiplied by an exponential term.

$$\mathcal{L}\left[f(t-T)\right] = e^{-sT} \mathcal{L}\left[f(t)\right] \tag{1.6}$$

To show this, apply the definition and change the integration variable so that $t' = t - T$

$$\int_0^\infty e^{-st} f(t-T)dt = \int_T^\infty e^{-s(t'+T)} f(t')dt' \tag{1.7}$$

After splitting the exponential term $e^{-s(t'+T)} = e^{-st'}e^{-sT}$, we see that we can take e^{-sT} outside of the integral because it does not depend on the integration variable.

$$\begin{aligned}\int_0^\infty e^{-st} f(t-T)dt &= \int_T^\infty e^{-s(t'+T)} f(t')dt' \\ &= e^{-sT}\int_T^\infty e^{-st'} f(t')dt' \\ &= e^{-sT}\int_0^\infty e^{-st'} f(t')dt' = e^{-sT}\mathcal{L}\left[f(t)\right]\end{aligned} \tag{1.8}$$

The integral that remains is just the Laplace transform of the unshifted function. The change of the lower integral bound in the last line (from T to 0) is allowed because the function is 0 for $t < 0$.

1.3.3 Derivation

The Laplace transform of the derivative of a function is expressed as

$$\mathcal{L}\left[\frac{df}{dt}\right] = s \cdot \mathcal{L}\left[f(t)\right] - f(0) \tag{1.9}$$

To show this, write the definition and integrate by parts:

$$\int_0^\infty \frac{df}{dt}e^{-st}dt = -\int_0^\infty (-s)f(t)e^{-st}dt + f(t)e^{-st}|_0^\infty = s\int_0^\infty f(t)e^{-st}dt - f(0) \tag{1.10}$$

1.3.4 Integration

The converse of the property of derivation comes into play when we integrate the signal over time:

$$\mathcal{L}\left[\int_0^t f(\tau)d\tau\right] = \frac{1}{s} \cdot \mathcal{L}\left[f(t)\right]. \tag{1.11}$$

This time, instead of multiplying by s, we are dividing by it. To show this, once again we start from the definition of the Laplace transform, then we integrate by parts:

$$\begin{aligned}&\int_0^\infty \left(\int_0^t f(\tau)d\tau\right) e^{-st}dt = \\ &\left[\left(\int_0^t f(\tau)d\tau\right)\left(-\frac{1}{s}\right)e^{-st}\right]_0^\infty - \int_0^\infty \frac{d}{dt}\left(\int_0^t f(\tau)d\tau\right)\left(-\frac{1}{s}\right)e^{-st}dt\end{aligned} \tag{1.12}$$

The left term is zero because when evaluated at infinity e^{-st} vanishes, and when evaluated at 0 the integral vanishes (because it because an integral between 0 and 0).

The right term contains the derivative of the integral, which is just the function itself:

$$\frac{d}{dt}\left(\int_0^t f(\tau)d\tau\right) = f(t). \tag{1.13}$$

Substituting back, we get

$$\int_0^\infty \left(\int_0^t f(\tau)d\tau\right) e^{-st}dt = \frac{1}{s}\int_0^\infty f(t)e^{-st}dt. \tag{1.14}$$

1.3.5 Convolution

Convolution is an operation between two signals $f(t)$ and $g(t)$. It gives us a new signal which in some sense is a combination of f and g, and in particular it is defined as follows:

$$f * g = \int_0^t f(t-\tau)g(\tau)d\tau. \tag{1.15}$$

One of the great advantages of using Laplace transforms is that it greatly simplifies computing convolutions. In fact, the convolution of two signals in the frequency domain is just a product of the Laplace transforms:

$$\mathcal{L}\left[f * g\right] = F(s)G(s) \tag{1.16}$$

The proof of this result is a bit more involved, and thus is omitted here.

1.3.6 Final value theorem

Given the Laplace transform $F(s)$ of some signal, the final value theorem tells us what is the behavior of the signal $f(t)$ as $t \to \infty$.

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} s \cdot F(s). \tag{1.17}$$

This theorem is especially important because it lets us determine the steady state behavior of a system in the time domain, given its description in the frequency domain. To show why the theorem is true, first recall that from the property of derivation we have

$$s\int_0^\infty f(t)e^{-st}dt = \int_0^\infty \frac{df}{dt}e^{-st}dt + f(0), \tag{1.18}$$

where the left side is just $sF(s)$. Now compute the limit of both sides for $s \to 0$. The exponential inside the integral simply becomes 1, and we are left with the integral of the derivative of f, which using the the fundamental theorem of calculus we can rewrite as

$$\int_0^\infty \frac{df}{dt}dt = \lim_{t\to\infty} f(t) - f(0). \tag{1.19}$$

Substituting back we get

$$\lim_{s\to 0} sF(s) = \lim_{t\to\infty} f(t) - f(0) + f(0) = \lim_{t\to\infty} f(t) \tag{1.20}$$

1.4 Laplace transform of simple functions

Function	$f(t)$	Laplace transform $F(s)$
Dirac delta	$\delta(t)$	1
Heaviside step function	$h(t)$	$\frac{1}{s}$
Ramp	t	$\frac{1}{s^2}$
Polynomial ramp	t^n	$\frac{n!}{s^{n+1}}$
Exponential	e^{at}	$\frac{1}{s-a}$
Sine	$\sin(\omega t)$	$\frac{\omega}{s^2+\omega^2}$
Cosine	$\cos(\omega t)$	$\frac{s}{s^2+\omega^2}$
Damped oscillation	$e^{-\sigma t}\sin(\omega t)$	$\frac{\omega}{(s+\sigma)^2+\omega^2}$
Damped oscillation	$e^{-\sigma t}\cos(\omega t)$	$\frac{s+\sigma}{(s+\sigma)^2+\omega^2}$

Table 1.1: Basic functions and their Laplace transforms.

Table 1.1 shows some simple functions and their Laplace transforms. This sections will show how to compute some of these results.

The simplest function is the impulse function, also known as the *Dirac delta* $\delta(t)$. This can be thought of as a function which is zero everywhere, except in $t = 0$, where it is infinite. It also has the property that its integral is equal to 1. Another property of the Dirac delta is that if multiplied by a function and then integrated, it returns the function sampled at zero. We can use this property to compute the Laplace transform.

$$X(s) = \int_0^\infty e^{-st}\delta(t)dt = 1 \tag{1.21}$$

The next example is the Heaviside step function. This is actually the integral of the Dirac delta.

$$h(t) = \begin{cases} 1 & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases} \tag{1.22}$$

Computing the Laplace transform of this function is fairly simple.

$$X(s) = \int_0^\infty e^{-st} h(t) dt = \int_0^\infty e^{-st} dt = \left[-\frac{1}{s} e^{-st} \right]_0^\infty = \frac{1}{s} \tag{1.23}$$

A ramp function is similarly defined, but instead of being constant it grows linearly

$$\rho(t) = \begin{cases} t & \text{if } t \geq 0 \\ 0 & \text{if } t < 0 \end{cases} \tag{1.24}$$

and its Laplace transform is

$$\int_0^\infty t e^{-st} dt = \left[-\frac{1}{s} t e^{-st} \right]_0^\infty - \int_0^\infty \left(-\frac{1}{s} \right) e^{-st} dt = \left[-\frac{1}{s^2} e^{-st} \right]_0^\infty = \frac{1}{s^2} \tag{1.25}$$

Exponential The exponential function $e^{at} h(t)$, which is multiplied by the Heaviside step function simply because we assume that all of our input signals are applied only starting at $t = 0$. The part before zero does not contribute to the integral, so we can safely ignore it. Computing the Laplace transform is then straightforward:

$$\mathcal{L}\left[e^{at} h(t) \right] = \int_0^\infty e^{-st} e^{at} dt = \int_0^\infty e^{(a-s)t} dt = \frac{1}{s-a} \tag{1.26}$$

Sine and Cosine The sine function is $\sin(\omega t) h(t)$.

$$\begin{aligned} \mathcal{L}\left[\sin(\omega t) h(t) \right] &= \int_0^\infty e^{-st} \sin(\omega t) dt \\ &= \left[e^{-st} \frac{1}{\omega} (-\cos(\omega t)) \right]_0^\infty - \int_0^\infty (-s) e^{-st} \frac{1}{\omega} (-\cos(\omega t)) dt \\ &= \frac{1}{\omega} - \frac{s}{\omega} \int_0^\infty e^{-st} \cos(\omega t) dt \\ &= \frac{1}{\omega} - \frac{s}{\omega} \left[e^{-st} \frac{1}{\omega} \sin(\omega t) \right]_0^\infty + \frac{s}{\omega} \int_0^\infty (-s) e^{-st} \frac{1}{\omega} \sin(\omega t) dt \\ &= \frac{1}{\omega} - \frac{s^2}{\omega^2} \int_0^\infty e^{-st} \sin(\omega t) dt = \frac{1}{\omega} - \frac{s^2}{\omega^2} \mathcal{L}\left[\sin(\omega t) h(t) \right] \end{aligned} \tag{1.27}$$

collect the common term on the left-hand side

$$\left(1 + \frac{s^2}{\omega^2} \right) \mathcal{L}\left[\sin(\omega t) h(t) \right] = \frac{1}{\omega} \tag{1.28}$$

from which we get

$$\mathcal{L}\left[\sin(\omega t) h(t) \right] = \frac{\omega}{s^2 + \omega^2} \tag{1.29}$$

The cosine function can be computed by simply using the derivation property of the Laplace transform:

$$\mathcal{L}\left[\cos(\omega t)\right] = \mathcal{L}\left[\frac{1}{\omega}\frac{d}{dt}\sin(\omega t)\right] = \frac{s}{\omega}\mathcal{L}\left[\sin(\omega t)\right] = \frac{s}{s^2+\omega^2} \tag{1.30}$$

Damped oscillations Damped oscillating functions are sine and cosine functions whose amplitude is modulated by a decreasing exponential

$$\mathcal{L}\left[e^{-\sigma t}\sin(\omega t)\right] = \int_0^\infty e^{-(s+\sigma)t}\sin(\omega t)dt = \frac{\omega}{(s+\sigma)^2+\omega^2} \tag{1.31}$$

which can be immediately derived from making the complex substitution $s' = s+\sigma$, and then making the inverse substitution to recover the original complex variable s. Similarly we find

$$\mathcal{L}\left[e^{-\sigma t}\cos(\omega t)\right] = \int_0^\infty e^{-(s+\sigma)t}\cos(\omega t)dt = \frac{s+\sigma}{(s+\sigma)^2+\omega^2} \tag{1.32}$$

1.4.1 Final Value Theorem Blackboard exercises

a system can subjected to a signal will modify it depending on its properties. Imagine that the system $F(s)$, described by the function in the Laplace domain $F(s) = \frac{10}{s^2+3s+2}$, is subjected to the step input $U(s) = \frac{1}{s}$ and we want to find the steady state value of the output signal $y(t)$ as $t \to \infty$

$$Y(s) = F(s)U(s) = \frac{10}{s(s^2+3s+2)}$$

we can apply the final value theorem for which

$$sY(s) = \frac{10s}{s(s^2+3s+2)} = \frac{10}{(s+1)(s+2)}$$

therefore

$$\lim_{t\to\infty} y(t) = \lim_{s\to 0} sY(s) = \frac{10}{(0+1)(0+2)} = 5$$

1.5 The transfer function

Consider a system to which we apply an input signal $U(s)$, and which outputs a signal $Y(s)$. The transfer function $W(s)$ represents the relationship between the input and the output in the frequency domain.

$$W(s) = \frac{Y(s)}{U(s)}. \tag{1.33}$$

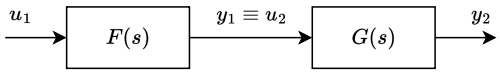


Figure 1.1: A series of transfer functions.

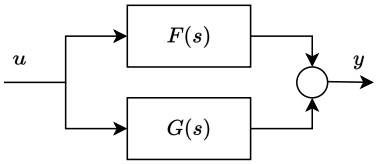


Figure 1.2: Two transfer functions in parallel.

1.5.1 Series of transfer functions

In a series of transfer function (Fig. 1.1), the output signal of the first is used as the input signal of the second:

$$\begin{aligned} Y_1(s) &= F(s)U_1(s) \\ Y_2(s) &= G(s)U_2(s) = G(s)Y_1(s) = G(s)F(s)U_1(s) \end{aligned} \tag{1.34}$$

Therefore, the transfer function of the system as a whole is

$$W(s) = \frac{Y_2(s)}{U_1(s)} = F(s)G(s) \tag{1.35}$$

In the time domain, this is equivalent to performing a convolution of the two signals.

1.5.2 Parallel of transfer functions

Having two transfer functions in parallel (Fig. 1.2) means that the input signal is applied as the input of both, and their outputs are summed, i.e.,

$$\begin{aligned} Y_1(s) &= F(s)U(s) \\ Y_2(s) &= G(s)U(s) \end{aligned} \tag{1.36}$$

Therefore, the transfer function of the system as a whole is

$$W(s) = \frac{Y_2(s)}{U_1(s)} = F(s) + G(s) \tag{1.37}$$

1.5.3 The closed-loop transfer function

Consider the block scheme of Fig. 1.3, which represents a closed-loop system with negative feedback. On the left-hand side of the diagram we have the input u, which is compared with output y by computing the difference. Let us name

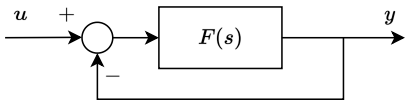


Figure 1.3: A closed loop system with negative feedback.

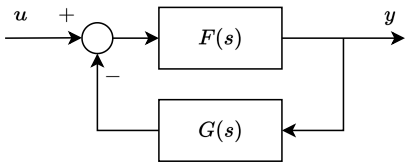


Figure 1.4: A closed loop system with negative feedback and a transfer function in the feedback loop.

this difference $e(t) = u(t) - y(t)$, as it represents the error. Its Laplace transform will be $E(s)$.

The output signal $Y(s)$ can be written in terms of the error signal $E(s)$ by simply multiplying the chain of transfer functions leading from the error to the output.

$$Y(s) = E(s)F(s), \tag{1.38}$$

but we can also express the error in terms of difference between input and output as $E(s) = U(s) - Y(s)$:

$$Y(s) = (U(s) - Y(s))F(s). \tag{1.39}$$

From this we can derive the transfer function of the closed-loop system:

$$\frac{Y(s)}{U(s)} = \frac{F(s)}{1 + F(s)}. \tag{1.40}$$

Fig. 1.4 also represents a feedback scheme, but here the feedback signal is passed through a block with transfer function $G(s)$. Also here we can start from expressing the output in terms of the error signal

$$Y(s) = E(s)F(s), \tag{1.41}$$

however, here the error must be computed as $E(s) = U(s) - G(s)Y(s)$. Therefore

$$\frac{Y(s)}{U(s)} = \frac{F(s)}{1 + G(s)F(s)}. \tag{1.42}$$

Chapter 2

Speed control

In this section we will discuss how to design a controller to achieve *speed control*. The goal of the speed control scheme is to track a given speed reference ω_{ref}, and it is shown in Fig. 2.1. The measured speed is compared with the reference, and their difference $\Delta\omega$ is used as the input to the *controller* block W_c.

The controller produces a *reference torque* τ_{ref}. This is the input to an *actuator* block $W_a(s)$, which produces the actual motor command to achieve such reference. The effective torque produced by the motor is the *electromotive torque* τ_e, to which we must subtract the torque applied from the load τ_l, which opposes the movement of the rotor. The difference of these two torque values enters the actual motor block $W_m(s)$, which represents the dynamics of the rotor. The output speed of the rotor ω is measured and used in the feedback loop to compute the speed error. However, it first goes through a block whose transfer function $W_s(s)$ represents the dynamics of the sensor performing the speed measurement.

In our initial analysis we will remove some components from this scheme in order to simplify the analysis. In particular, we will assume that both the dynamics of the sensor $W_s(s)$ and the actuator $W_a(s)$ are significantly faster than the dynamics of the motor $W_m(s)$, and therefore we can approximate then as $W_s(s) = W_a(s) = 1$. The resulting scheme is shown in Fig. 2.2.

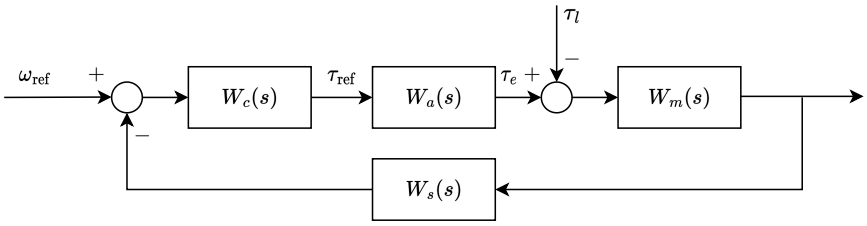


Figure 2.1: Block scheme of a speed controller.

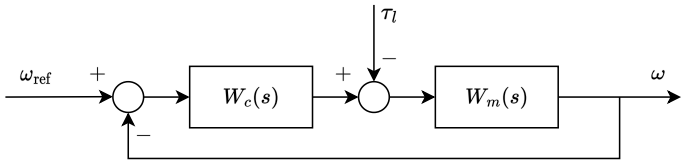


Figure 2.2: Simplified block scheme of a speed controller.

2.0.1 Motor model and transfer function

The dynamics of the rotor can be written as

$$J\frac{d\omega}{dt} = \tau_e - \tau_l - B\omega, \tag{2.1}$$

where J is the moment of inertia and ω is the speed of the rotor. $B\omega$ is the effect of friction.

To derive the transfer function of the motor, first perform a Laplace transform operation on both sides

$$Js\Omega = T_e - T_l - B\Omega, \tag{2.2}$$

where now $T_e(s)$, $T_l(s)$ and $\Omega(s)$ are the transforms of τ_e, τ_l and ω, and we have applied the derivation property to the left side by multiplying by s. The transfer function of this block can be computed by noting that the output is Ω and the input is the difference $T_e - T_l$:

$$W_m(s) = \frac{\Omega}{T_e - T_l} = \frac{1}{Js + B}. \tag{2.3}$$

If we neglect friction and have no load torque, then the transfer function becomes

$$W_m(s) = \frac{1}{Js} \tag{2.4}$$

which is essentially an integrator.

2.1 Proportional feedback action

We will now see the simplest speed control scheme, which is the proportional controller. The action is proportional to the speed error $\Delta\omega$.

$$\tau_{\text{ref}} = k_p \cdot \Delta\omega \tag{2.5}$$

The corresponding controller transfer function is simply a constant gain:

$$W_c(s) = \frac{T_{\text{ref}}(s)}{\Omega(s)} = k_p \tag{2.6}$$

2.1.1 No load, no friction

The transfer function of the closed-loop system is

$$W(s) = \frac{W_c(s)W_m(s)}{1 + W_c(s)W_m(s)} = \frac{k_p/(Js)}{1 + k_p/(Js)} = \frac{k_p}{J}\frac{1}{s + k_p/J}. \tag{2.7}$$

We are interested in determining the steady-state behavior of this system, i.e., whether or not ω will reach the target value ω_{ref} for $t \to \infty$.

If we apply a constant $\bar{\omega}_{\text{ref}}$ starting at time $t = 0$, this corresponds to a step function $\bar{\omega}_{\text{ref}}h(t)$, and its Laplace transform is

$$\Omega_{\text{ref}}(s) = \frac{\bar{\omega}_{\text{ref}}}{s}. \tag{2.8}$$

The output signal is then

$$\Omega(s) = \Omega_{\text{ref}}(s)W(s) = \frac{\bar{\omega}_{\text{ref}}}{s}\frac{k_p}{J}\frac{1}{s + k_p/J}. \tag{2.9}$$

Now we perform a partial fraction decomposition:

$$\frac{\bar{\omega}_{\text{ref}}k_p}{J}\frac{1}{s(s + k_p/J)} = \frac{\bar{\omega}_{\text{ref}}k_p}{J}\left(\frac{A_1}{s} + \frac{A_2}{s + k_p/J}\right) = \frac{\bar{\omega}_{\text{ref}}k_p}{J}\frac{(A_1 + A_2)s + A_1k_p/J}{s(s + k_p/J)}, \tag{2.10}$$

from which we find $A_1 + A_2 = 0$ and $A_1 = J/k_p$, thus

$$\Omega(s) = \frac{\bar{\omega}_{\text{ref}}k_p}{J}\left(\frac{J/k_p}{s} + \frac{-J/k_p}{s + k_p/J}\right) = \bar{\omega}_{\text{ref}}\left(\frac{1}{s} - \frac{1}{s + k_p/J}\right). \tag{2.11}$$

To go back to the time domain, we must find the inverse transform of this expression, which we can do simply by looking to Table 1.1 in reverse. The first term corresponds to a step function, and the second term corresponds to an exponential.

$$\omega(t) = \bar{\omega}_{\text{ref}}h(t)\left(1 - e^{-\frac{k_p}{J}t}\right) \tag{2.12}$$

In this case it is easy to see the long term behavior because the resulting expression is quite simple. Let us nevertheless approach it in a more systematic way, and determine the steady-state behavior using the final value theorem:

$$\lim_{t\to\infty}\omega(t) = \lim_{s\to 0}sW_s(s)\Omega_{\text{ref}}(s) = \lim_{s\to 0}s\frac{\bar{\omega}_{\text{ref}}k_p}{J}\frac{1}{s(s + k_p/J)} = \bar{\omega}_{\text{ref}}, \tag{2.13}$$

which indicates that the speed actually converges to the desired value, regardless of the chosen gain. Still, keep in mind that here we neglected the friction term and we assumed that there is no load torque.

2.1.2 Effect of load

A load acts as a disturbance. The effect using a proportional controller can be found by exploiting the property of linearity: we compute separately the effect of the reference and the effect of the load, then sum the outputs.

The input-output transfer function $W_{\mathrm{i/o}}$ can be computed by setting the load to zero:

$$\Omega = W_m T_e = W_m W_c \Delta\Omega = W_m W_c (\Omega_{\mathrm{ref}} - \Omega) \tag{2.14}$$

$$W_{\mathrm{i/o}} = \frac{\Omega}{\Omega_{\mathrm{ref}}} = \frac{W_m W_c}{1 + W_m W_c}. \tag{2.15}$$

This is effectively what we computed in the previous case using the closed loop transfer function.

To compute the disturbance-output transfer function $W_{\mathrm{d/o}}$, we set the reference to zero, then start from the output and follow the block scheme in reverse, just as we did for the input-output transfer function.

$$\Omega = W_m(T_e - T_l) = W_m(W_c \Delta\Omega - T_l) = W_m(-W_c \Omega - T_l) \tag{2.16}$$

$$W_{\mathrm{d/o}} = \frac{\Omega}{T_l} = -\frac{W_m}{1 + W_m W_c} \tag{2.17}$$

We can now sum the two contributions to find the output under the effect of both the reference and the disturbance:

$$\Omega(s) = W_{\mathrm{i/o}} \Omega_{\mathrm{ref}} + W_{\mathrm{d/o}} T_l = \frac{W_m W_c}{1 + W_m W_c} \Omega_{\mathrm{ref}} - \frac{W_m}{1 + W_m W_c} T_l \tag{2.18}$$

Now we substitute $W_m = 1/(Js)$ for the motor and $W_c = k_p$ for the controller.

$$\begin{aligned} \Omega(s) &= \frac{k_p/(Js)}{1 + k_p/(Js)} \Omega_{\mathrm{ref}} - \frac{1/(Js)}{1 + k_p/(Js)} T_l \\ &= \frac{k_p}{J} \frac{1}{s + k_p/J} \Omega_{\mathrm{ref}} - \frac{1}{J} \frac{1}{s + k_p/J} T_l \\ &= \frac{k_p}{J} \frac{1}{s + k_p/J} \left(\Omega_{\mathrm{ref}} - \frac{1}{k_p} T_l \right) \end{aligned} \tag{2.19}$$

From this, we can see that the time-domain evolution of the output is

$$\omega(t) = \left(\bar{\omega}_{\mathrm{ref}} - \frac{\tau_l}{k_p} \right) \left(1 - e^{-\frac{k_p}{J} t} \right) \tag{2.20}$$

and thus that the output does not tend to the desired value, but rather

$$\lim_{t \to \infty} \omega(t) = \bar{\omega}_{\mathrm{ref}} - \frac{\tau_l}{k_p}, \tag{2.21}$$

i.e., there is a *steady-state error*, that can be reduced by increasing the controller gain k_p. This can be seen by computing the limit of the time expression, but also with the final value theorem:

$$\begin{aligned}\lim_{t\to\infty}\omega(t) &= \lim_{s\to 0} s\cdot\Omega(s)\\ &= \lim_{s\to 0} s\cdot\frac{k_p}{J}\frac{1}{s+k_p/J}\left(\Omega_{\text{ref}}-\frac{1}{k_p}T_l\right)\\ &= \lim_{s\to 0} s\cdot\frac{k_p}{J}\frac{1}{s+k_p/J}\left(\frac{\bar{\omega}_{\text{ref}}}{s}-\frac{1}{k_p}\frac{\tau_l}{s}\right)\\ &= \bar{\omega}_{\text{ref}}-\frac{\tau_l}{k_p}\end{aligned} \tag{2.22}$$

2.1.3 Variable reference

Up until now, we have assumed that our reference was given by a step function, which implies that we are trying to reach a constant speed value and stay there. This is known as a *regulation* problem, i.e., the reference is a constant value of the output.

Now we will analyze what happens if the goal is to follow a reference that varies in time. This is known as a *servo* problem.

Let us go back to the case in which no load is applied (and we still have no friction). We already computed the transfer function of the closed loop system under proportional action in Eq. (2.9):

$$\Omega(s) = W_{\text{i/o}}(s)\cdot\Omega_{\text{ref}}(s) = \frac{k_p}{J}\frac{1}{s+k_p/J}\cdot\Omega_{\text{ref}}(s). \tag{2.23}$$

Now, instead of substituting the Ω_{ref} of the step function, we would like to see what happens if the reference is given by a ramp function, i.e.,

$$\Omega_{\text{ref}}(s) = \frac{\alpha_{\text{ref}}}{s^2}. \tag{2.24}$$

The output is

$$\Omega(s) = \frac{k_p}{J}\frac{1}{s+k_p/J}\cdot\frac{\alpha_{\text{ref}}}{s^2}. \tag{2.25}$$

If we compute long-term behavior using the final value theorem, we get

$$\lim_{t\to\infty}\omega(t) = \lim_{s\to 0} s\cdot\Omega(s) = \infty \tag{2.26}$$

because there is an s in the denominator that does not get simplified. However, this is not surprising, because the reference itself was a signal that grows to infinity. In fact, analyzing the long term behavior of the output in this case is simply not informative.

What we can do is instead analyze the *error* signal $\Delta\omega(t) = \omega_{\text{ref}}(t) - \omega(t)$. In the frequency domain this is

$$\Delta\Omega(s) = \Omega_{\text{ref}}(s) - \Omega(s) = (1 - W\text{i/o}(s))\Omega_{\text{ref}}(s) \tag{2.27}$$

For the case of a ramp function, we have

$$\Delta\Omega(s) = \left(1 - \frac{k_p}{J}\frac{1}{s + k_p/J}\right)\frac{\alpha_{\text{ref}}}{s^2} = \frac{s}{s + k_p/J}\frac{\alpha_{\text{ref}}}{s^2} \tag{2.28}$$

If we apply the final value theorem to find the steady-state value of the error, we get

$$\begin{aligned}\lim_{t\to\infty} \Delta\omega(t) &= \lim_{s\to 0} s \cdot \Delta\Omega(s) \\ &= \lim_{s\to 0} s\frac{s}{s + k_p/J}\frac{\alpha_{\text{ref}}}{s^2} \\ &= \frac{J}{k_p}\alpha_{\text{ref}}.\end{aligned} \tag{2.29}$$

We can see that there is a non-zero steady-state error, which gets reduced by increasing the gain k_p.

2.1.4 Effect of friction

Up until now we have neglected the effect of friction. If instead we take it into account, we must go back to the motor model

$$W_m(s) = \frac{1}{Js + B} \tag{2.30}$$

Let us see how our proportional control scheme behaves under these circumstances. The closed-loop input-output transfer function is

$$\begin{aligned}W_{\text{i/o}}(s) &= \frac{W_c(s)W_m(s)}{1 + W_c(s)W_m(s)} \\ &= \frac{k_p/(Js + B)}{1 + k_p/(Js + B)} \\ &= \frac{k_p}{Js + B + k_p}\end{aligned} \tag{2.31}$$

The input-error transfer function is

$$\begin{aligned}W_{\text{i/e}}(s) &= 1 - W_{\text{in}}(s) \\ &= 1 - \frac{k_p}{Js + B + k_p} \\ &= \frac{Js + B}{Js + B + k_p}.\end{aligned} \tag{2.32}$$

We can now write the error in response to a step function as

$$\Delta\Omega(s) = W_{\text{err}}(s)\frac{\omega_{\text{ref}}}{s} \tag{2.33}$$

from which we can determine the steady-state behavior using the final value theorem:

$$\begin{aligned}\lim_{t\to\infty}\Delta\omega(t) &= \lim_{s\to 0} s\cdot\Delta\Omega(s)\\ &= \lim_{s\to 0} s\cdot W_{\text{err}}(s)\frac{\omega_{\text{ref}}}{s}\\ &= \lim_{s\to 0} s\frac{Js+B}{Js+B+k_p}\frac{\omega_{\text{ref}}}{s}\\ &= \frac{B}{B+k_p}\omega_{\text{ref}}.\end{aligned} \tag{2.34}$$

We notice a few things:

- for $B=0$ we recover the case with no friction, as the error is 0;
- for $B>0$ we get a steady-state error, as $B/(B+k_p)$ is between 0 and 1 (because both B and k_p are positive numbers);
- if we wish to reduce the error, we can increase the gain: once again we see that for $k_p\to\infty$, the error tends to zero.

2.2 Speed control: Proportional + integral action

This controller combines a proportional action analogous to the one we used up until now, and an integral action, which as the name suggests is proportional to the integral over time of the speed error.

$$\tau_e(t) = k_p\Delta\omega(t) + k_i\int_0^t \Delta\omega(\tau)d\tau \tag{2.35}$$

In the Laplace transform of this signal is

$$T_e(s) = k_p\Delta\Omega(s) + \frac{k_i}{s}\Delta\Omega(s) = \left(k_p+\frac{k_i}{s}\right)\Delta\Omega(s) \tag{2.36}$$

which means that the controller transfer function is

$$W_c(s) = \frac{T_e(s)}{\Delta\Omega(s)} = k_p + \frac{k_i}{s} \tag{2.37}$$

Let us now see how this controller behaves in the scenarios we saw for the proportional controller.

2.2.1 No load, no friction

We will first compute the input-output transfer function, recalling that with no friction the motor transfer function is $1/(Js)$, and the proportional-integral controller transfer function is $k_p + k_i/s$, we get

$$\begin{aligned} W_{\mathrm{i/o}}(s) &= \frac{W_c(s)W_m(s)}{1 + W_c(s)W_m(s)} \\ &= \frac{(k_p + k_i/s)/(Js)}{1 + (k_p + k_i/s)/(Js)} \\ &= \frac{k_p s + k_i}{Js^2 + k_p s + k_i}. \end{aligned} \tag{2.38}$$

The input-error transfer function is

$$\begin{aligned} W_{\mathrm{i/e}}(s) &= 1 - W_{\mathrm{i/o}}(s) \\ &= 1 - \frac{k_p s + k_i}{Js^2 + k_p s + k_i} \\ &= \frac{Js^2}{Js^2 + k_p s + k_i} \end{aligned} \tag{2.39}$$

Using the final value theorem, we can determine the steady-state error when the reference is given by a step function:

$$\begin{aligned} \lim_{t\to\infty} \Delta\omega(t) &= \lim_{s\to 0} s \cdot \Delta\Omega(s) \\ &= \lim_{s\to 0} s \cdot W_{\mathrm{i/e}}(s) \cdot \frac{\bar{\omega}_{\mathrm{ref}}}{s} \\ &= \lim_{s\to 0} s \frac{Js^2}{Js^2 + k_p s + k_i} \frac{\bar{\omega}_{\mathrm{ref}}}{s} = 0. \end{aligned} \tag{2.40}$$

We find that the steady state error is zero, which is to be expected because we had already achieved this using the proportional controller alone.

2.2.2 Load rejection

Let us now add a load, and see how well this controller behaves in its presence. As we have already seen, thanks to the linearity of the Laplace transform, we can compute separately the effects of applying an input reference and a load, and sum the resulting contributions. We have already computed the contribution of the input reference, it is the transfer function $W_{\mathrm{i/o}}(s)$. Now we need to compute the transfer function between the disturbance (the load) and the output.

$$\begin{aligned} W_{\mathrm{d/o}}(s) &= -\frac{W_m(s)}{1 + W_c(s)W_m(s)} \\ &= -\frac{1/(Js)}{1 + (k_p + k_i/s)/(Js)} \\ &= -\frac{s}{Js^2 + k_p s + k_i}. \end{aligned} \tag{2.41}$$

Now the output is

$$\Omega(s) = W_{\mathrm{i/o}}(s)\Omega_{\mathrm{ref}}(s) - W_{\mathrm{d/o}}(s)T_l(s). \tag{2.42}$$

If both the reference and the load are step functions, i.e., $\Omega_{\mathrm{ref}}(s) = \bar{\omega}_{\mathrm{ref}}/s$ and $T_l(s) = \tau_l/s$, we have

$$\Omega(s) = \frac{k_p s + k_i}{Js^2 + k_p s + k_i}\frac{\bar{\omega}_{\mathrm{ref}}}{s} - \frac{s}{Js^2 + k_p s + k_i}\frac{\tau_l}{s} \tag{2.43}$$

And the error is

$$\begin{aligned}
\Delta\Omega(s) &= \Omega_{\mathrm{ref}}(s) - \Omega(s) \\
&= \frac{\bar{\omega}_{\mathrm{ref}}}{s} - \frac{k_p s + k_i}{Js^2 + k_p s + k_i}\frac{\bar{\omega}_{\mathrm{ref}}}{s} + \frac{s}{Js^2 + k_p s + k_i}\frac{\tau_l}{s} \\
&= \left(1 - \frac{k_p s + k_i}{Js^2 + k_p s + k_i}\right)\frac{\bar{\omega}_{\mathrm{ref}}}{s} + \frac{s}{Js^2 + k_p s + k_i}\frac{\tau_l}{s} \\
&= \frac{Js^2}{Js^2 + k_p s + k_i}\frac{\bar{\omega}_{\mathrm{ref}}}{s} + \frac{s}{Js^2 + k_p s + k_i}\frac{\tau_l}{s} \\
&= \frac{J\bar{\omega}_{\mathrm{ref}} s + \tau_l}{Js^2 + k_p s + k_i}
\end{aligned} \tag{2.44}$$

Let us now use the final value theorem to find the steady-state error:

$$\begin{aligned}
\lim_{t\to\infty} \Delta\omega(t) &= \lim_{s\to 0} s \cdot \Delta\Omega(s) \\
&= \lim_{s\to 0} s\frac{J\bar{\omega}_{\mathrm{ref}} s + \tau_l}{Js^2 + k_p s + k_i} = 0.
\end{aligned} \tag{2.45}$$

As we can see, the PI controller is able to completely reject the effect of a constant load at steady-state.

2.2.3 Load and friction

Consider now the case in which we have both load and friction. The motor model is $1/(Js + B)$. Let us compute the input-output transfer function:

$$\begin{aligned}
W_{\mathrm{i/o}}(s) &= \frac{W_c(s)W_m(s)}{1 + W_c(s)W_m(s)} \\
&= \frac{(k_p + k_i/s)/(Js + B)}{1 + (k_p + k_i/s)/(Js + B)} \\
&= \frac{k_p s + k_i}{Js^2 + (B + k_p)s + k_i}
\end{aligned} \tag{2.46}$$

and the disturbance-output transfer function:

$$\begin{aligned} W_{\mathrm{d/o}}(s) &= -\frac{W_m(s)}{1 + W_c(s)W_m(s)} \\ &= -\frac{1/(Js+B)}{1 + (k_p + k_i/s)/(Js+B)} \\ &= -\frac{s}{Js^2 + (B+k_p)s + k_i} \end{aligned} \tag{2.47}$$

Thus the output signal is

$$\Omega(s) = W_{\mathrm{i/o}}(s)\Omega_{\mathrm{ref}}(s) - W_{\mathrm{d/o}}(s)T_l(s). \tag{2.48}$$

If both the reference and the load are step functions, i.e., $\Omega_{\mathrm{ref}}(s) = \bar{\omega}_{\mathrm{ref}}/s$ and $T_l(s) = \tau_l/s$, we have

$$\Omega(s) = \frac{k_p s + k_i}{Js^2 + (B+k_p)s + k_i}\frac{\bar{\omega}_{\mathrm{ref}}}{s} - \frac{s}{Js^2 + (B+k_p)s + k_i}\frac{\tau_l}{s} \tag{2.49}$$

And the error is

$$\begin{aligned} \Delta\Omega(s) &= \Omega_{\mathrm{ref}}(s) - \Omega(s) \\ &= \frac{\bar{\omega}_{\mathrm{ref}}}{s} - \frac{k_p s + k_i}{Js^2 + (B+k_p)s + k_i}\frac{\bar{\omega}_{\mathrm{ref}}}{s} + \frac{s}{Js^2 + (B+k_p)s + k_i}\frac{\tau_l}{s} \\ &= \left(1 - \frac{k_p s + k_i}{Js^2 + (B+k_p)s + k_i}\right)\frac{\bar{\omega}_{\mathrm{ref}}}{s} + \frac{s}{Js^2 + (B+k_p)s + k_i}\frac{\tau_l}{s} \\ &= \frac{Js^2 + Bs}{Js^2 + (B+k_p)s + k_i}\frac{\bar{\omega}_{\mathrm{ref}}}{s} + \frac{s}{Js^2 + (B+k_p)s + k_i}\frac{\tau_l}{s} \\ &= \frac{J\bar{\omega}_{\mathrm{ref}}s + B + \tau_l}{Js^2 + (B+k_p)s + k_i} \end{aligned} \tag{2.50}$$

Let us now use the final value theorem to find the steady-state error:

$$\begin{aligned} \lim_{t\to\infty} \Delta\omega(t) &= \lim_{s\to 0} s \cdot \Delta\Omega(s) \\ &= \lim_{s\to 0} s\frac{J\bar{\omega}_{\mathrm{ref}}s + B + \tau_l}{Js^2 + k_p s + k_i} = 0. \end{aligned} \tag{2.51}$$

Even with the added friction, the PI controller reaches zero steady-state error for a constant reference.

2.2.4 PI controller under variable reference

Consider once again the output signal for a PI-controller system:

$$\Omega(s) = W_{\mathrm{i/o}}(s)\Omega_{\mathrm{ref}}(s) - W_{\mathrm{d/o}}(s)T_l(s). \tag{2.52}$$

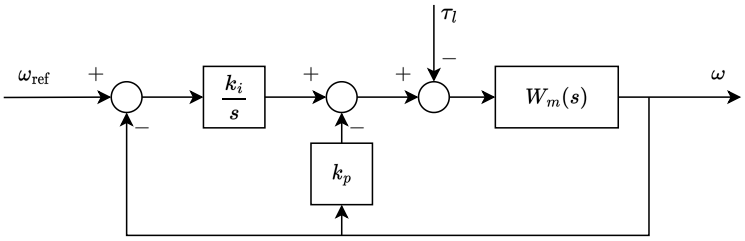


Figure 2.3: Block scheme of the relocated proportional speed controller.

If now we pick a ramp function for the reference, i.e., $\Omega_{\text{ref}} = \alpha_{\text{ref}}/s^2$, still keeping the load as a step function, we obtain

$$\Omega(s) = \frac{k_p s + k_i}{Js^2 + (B + k_p)s + k_i}\frac{\alpha_{\text{ref}}}{s^2} - \frac{s}{Js^2 + (B + k_p)s + k_i}\frac{\tau_l}{s} \tag{2.53}$$

The error signal is

$$\begin{aligned}
\Delta\Omega(s) &= \Omega_{\text{ref}}(s) - \Omega(s) \\
&= \frac{\alpha_{\text{ref}}}{s^2} - \frac{k_p s + k_i}{Js^2 + (B + k_p)s + k_i}\frac{\alpha_{\text{ref}}}{s^2} + \frac{s}{Js^2 + (B + k_p)s + k_i}\frac{\tau_l}{s} \\
&= \left(1 - \frac{k_p s + k_i}{Js^2 + (B + k_p)s + k_i}\right)\frac{\alpha_{\text{ref}}}{s^2} + \frac{s}{Js^2 + (B + k_p)s + k_i}\frac{\tau_l}{s} \\
&= \frac{Js + B}{Js^2 + (B + k_p)s + k_i}\frac{\alpha_{\text{ref}}}{s} + \frac{1}{Js^2 + (B + k_p)s + k_i}\tau_l
\end{aligned} \tag{2.54}$$

Without further manipulations, let us just apply the final value theorem to find the long-term behavior of this system:

$$\begin{aligned}
\lim_{t\to\infty} \Delta\omega(t) &= \lim_{s\to 0} s \cdot \Delta\Omega(s) \\
&= \lim_{s\to 0} s\left(\frac{Js + B}{Js^2 + (B + k_p)s + k_i}\frac{\alpha_{\text{ref}}}{s} + \frac{1}{Js^2 + (B + k_p)s + k_i}\tau_l\right) \\
&= \lim_{s\to 0} \frac{Js + B}{Js^2 + (B + k_p)s + k_i}\alpha_{\text{ref}} = \frac{B}{k_i}\alpha_{\text{ref}}
\end{aligned} \tag{2.55}$$

So, the load plays no effect, because the steady-state value is independent of τ_l, but the ramp reference still results in a steady-state non-zero error, which can be reduced by increasing the integral gain k_i.

2.2.5 Relocated proportional action

Using a relocated proportional action scheme, we get the output signal

$$\begin{aligned}\Omega(s) &= W_m(s)\left(T_e(s) - T_l(s)\right) \\ &= W_m(s)\left(\frac{k_i}{s}\Delta\Omega(s) - k_p\Omega(s) - T_l(s)\right) \\ &= \frac{1}{Js+B}\left(\frac{k_i}{s}\Omega_{\text{ref}}(s) - \left(k_p + \frac{k_i}{s}\right)\Omega(s) - T_l(s)\right)\end{aligned} \tag{2.56}$$

We can then determine the input-output transfer function

$$\begin{aligned}W_{\text{i/o}}(s) = \left.\frac{\Omega(s)}{\Omega_{\text{ref}}}\right|_{T_l=0} &= \\ &= \frac{(k_i/s)/(Js+B)}{1+(k_p+k_i/s)/(Js+B)} \\ &= \frac{k_i}{Js^2+(B+k_p)s+k_i}\end{aligned} \tag{2.57}$$

which has the same denominator of the transfer function for the standard PI scheme, but it has no zeros in the denominator. It is easy to verify that the disturbance-output transfer function is identical to that of the standard PI controller.

The relocated scheme has a similar performance to the standard PI scheme, in terms of load rejection. It is capable of achieving zero steady-state error in response to a constant load disturbance. However, the transient characteristics are different due to the absence of the closed loop zero. In particular, the relocated scheme has a slower response.

Chapter 3

Transient analysis

Both the proportional-integral schemes (the standard one and the relocated one) have a transfer function of the following form:

$$W_{\mathrm{i/o}}(s) = \frac{N(s)}{D(s)} = \frac{N(s)}{s^2 + \frac{B+k_p}{J}s + \frac{k_i}{J}}. \tag{3.1}$$

This is a *second order system*, because the degree of the polynomial in the denominator is 2. It is common to rewrite the denominator polynomial in terms of natural pulse ω_n and damping ζ:

$$\omega_n = \sqrt{\frac{k_i}{J}}, \quad \zeta = \frac{k_p + B}{2J\omega_n} \tag{3.2}$$

$$W_{\mathrm{i/o}}(s) = \frac{N(s)}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{3.3}$$

The denominator written in this form can be factored to find its roots. These are the transfer function *poles*, and they turn out to be

$$\begin{aligned} s_{1,2} &= \frac{-2\zeta\omega_n \pm \sqrt{4\zeta^2\omega_n^2 - 4\omega_n^2}}{2} \\ &= -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \end{aligned} \tag{3.4}$$

There can be one of three cases, depending on the value of ζ:

- $\zeta > 1$: we have two distinct real poles $s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$
- $\zeta = 0$: we have two coincident real poles $s_{1,2} = -\zeta\omega_n$
- $\zeta < 0$: we have two conjugate complex poles $s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$

3.1 Second order step response

For the moment, let's consider the third case, because it will lead to the most interesting solutions. From eq. (3.4), we can rename the real and imaginary part of these poles to

$$\begin{aligned} \sigma &= \zeta\omega_n \\ \omega_0 &= \omega_n\sqrt{1-\zeta^2} \end{aligned} \tag{3.5}$$

and we write them as

$$s_{1,2} = -\sigma \pm j\omega_0 \tag{3.6}$$

In the following it will be useful to remember that $s_1 s_2 = \omega_n^2$ and $s_1 + s_2 = -2\sigma$.

Consider now the scheme with the relocated proportional action. We found its transfer function to be

$$W_{\mathrm{i/o}}(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s+\sigma+j\omega_0)(s+\sigma-j\omega_0)} \tag{3.7}$$

which we can rewrite as

$$W_{\mathrm{i/o}}(s) = \frac{\omega_n^2}{(s+\sigma)^2 + \omega_0^2} \tag{3.8}$$

If we apply a step input we get the output response

$$\Omega(s) = \frac{\omega_n^2}{(s+\sigma)^2 + \omega_0^2}\frac{\bar{\omega}_{\mathrm{ref}}}{s} \tag{3.9}$$

Let's now try to separate this into separate terms

$$\begin{aligned} \Omega(s) &= \bar{\omega}_{\mathrm{ref}}\left(\frac{A_1}{s} + \frac{A_2 s + A_3}{(s+\sigma)^2 + \omega_0^2}\right) \\ &= \bar{\omega}_{\mathrm{ref}}\frac{A_1((s+\sigma)^2 + \omega_0^2) + (A_2 s + A_3)s}{s((s+\sigma)^2 + \omega_0^2)} \end{aligned} \tag{3.10}$$

Note that the term multiplying A_1 in the numerator is just a different way of rewriting the denominator of $W_{\mathrm{i/o}}(s)$. For convenience, expanding it and substituting the relations in eq. (3.5) we get that:

$$\begin{aligned} \Omega(s) &= \bar{\omega}_{\mathrm{ref}}\frac{A_1(s^2 + 2\sigma s + \omega_n^2) + A_2 s^2 + A_3 s}{s((s+\sigma)^2 + \omega_0^2)} \\ &= \bar{\omega}_{\mathrm{ref}}\frac{(A_1 + A_2)s^2 + (2A_1\sigma + A_3)s + \omega_n^2}{s((s+\sigma)^2 + A_1\omega_0^2)} \end{aligned} \tag{3.11}$$

We get the following conditions on A_1, A_2 and A_3:

$$\begin{aligned} A_1 + A_2 &= 0 \\ 2A_1\sigma + A_3 &= 0 \\ A_1\omega_n^2 &= \omega_n^2 \end{aligned} \tag{3.12}$$

which we can easily solve to find

$$\begin{aligned} A_1 &= 1 \\ A_2 &= -1 \\ A_3 &= -2\sigma \end{aligned} \tag{3.13}$$

And thus we have factored the expression as

$$\begin{aligned} \Omega(s) &= \bar{\omega}_{\text{ref}} \left(\frac{1}{s} - \frac{s+2\sigma}{(s+\sigma)^2 + \omega_0^2} \right) \\ &= \bar{\omega}_{\text{ref}} \left(\frac{1}{s} - \frac{s+\sigma}{(s+\sigma)^2 + \omega_0^2} - \frac{\sigma}{(s+\sigma)^2 + \omega_0^2} \right) \\ &= \bar{\omega}_{\text{ref}} \left(\frac{1}{s} - \frac{s+\sigma}{(s+\sigma)^2 + \omega_0^2} - \frac{\sigma}{\omega_0} \frac{\omega_0}{(s+\sigma)^2 + \omega_0^2} \right) \end{aligned} \tag{3.14}$$

The inverse transform of this expression is

$$\begin{aligned} \omega(t) &= \bar{\omega}_{\text{ref}} \left(1 - e^{-\sigma t} \cos(\omega_0 t) - \frac{\sigma}{\omega_0} e^{-\sigma t} \sin(\omega_0 t) \right) \\ &= \bar{\omega}_{\text{ref}} \left(1 - e^{-\sigma t} \cos(\omega_0 t) - \frac{\sigma}{\omega_0} e^{-\sigma t} \sin(\omega_0 t) \right) \\ &= \bar{\omega}_{\text{ref}} \left(1 - e^{-\sigma t} \cos(\omega_0 t) - \frac{\zeta}{\sqrt{1-\zeta^2}} e^{-\sigma t} \sin(\omega_0 t) \right) \\ &= \bar{\omega}_{\text{ref}} \left(1 - \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}} \left(\sqrt{1-\zeta^2} \cos(\omega_0 t) - \zeta \sin(\omega_0 t) \right) \right) \end{aligned} \tag{3.15}$$

In the above expression, if we introduce a *phase* ϕ, such that $\zeta = \cos\phi$, we can transform it into

$$\boxed{\omega(t) = \bar{\omega}_{\text{ref}} \left(1 - \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}} \sin(\omega_0 t + \phi) \right)} \tag{3.16}$$

This is the time expression of the step response of a second order system with complex conjugate poles. It is essentially an oscillating signal, whose amplitude decreases by virtue of the exponential term $e^{-\sigma t}$.

Note that introducing ϕ such that $\zeta = \cos\phi$ does not replace or alter the dynamic dependence of ζ. It is a standard trigonometric substitution used to simplify the sum of sinusoids $\sqrt{1-\zeta^2}\cos(\omega_0 t) - \zeta \sin(\omega_0 t)$. Since the system is underdamped, $\zeta \in (0, 1)$, guaranteeing that ζ falls in the valid range $[-1, 1]$ for the cosine function. Defining an angle $\phi \in (0, \pi/2)$ such taht

$$\begin{aligned} \cos\phi &= \zeta \\ \sin\phi &= \sqrt{1-\zeta^2} \end{aligned} \tag{3.17}$$

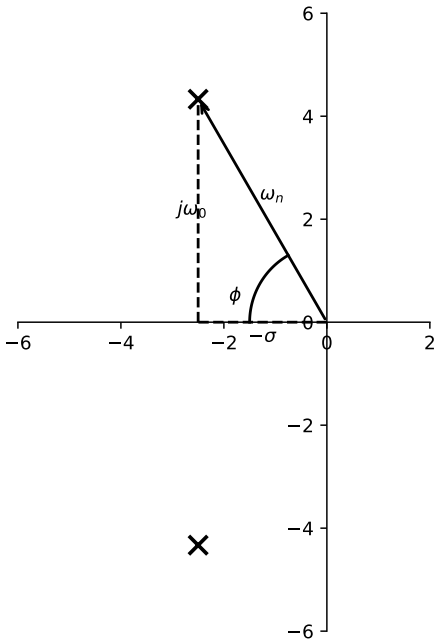


Figure 3.1: s-plane Geometry of the complex poles

transforms the bracketed expression via the angle addition identity for which

$$\sin\phi\cos(\omega_0 t) + \cos\phi\sin(\omega_0 t) = \sin(\omega_0 + \phi) \tag{3.18}$$

Geometrically in the complex s-plane, the poles are located at $s_{1,2} = -\sigma \pm j\omega_0$. The distance from the origin to the pole is $\sqrt{\sigma^2 + \omega_0^2} = \omega_n$ Therefore, $\cos\phi = \frac{\sigma}{\omega_n} = \zeta$, meaning ϕ is the angle between the negative real axis and the pole vector in the complex plane (see fig.(3.1)).

3.2 Transient characteristics

For the relocated scheme, which has no zeros, the response can be seen in Fig. 3.2. In particular, we can see three different types of response, that can be characterized in one of the following ways

- $\zeta < 1$: underdamped, the step response presents oscillations
- $\zeta = 1$: critical, relatively fast response, no oscillations
- $\zeta > 1$: overdamped, no oscillations and slower response

One important aspect of the response for $\zeta \geq 1$ is that no *overshoot* is present, i.e., the response never rises above the reference signal. Note that this is true

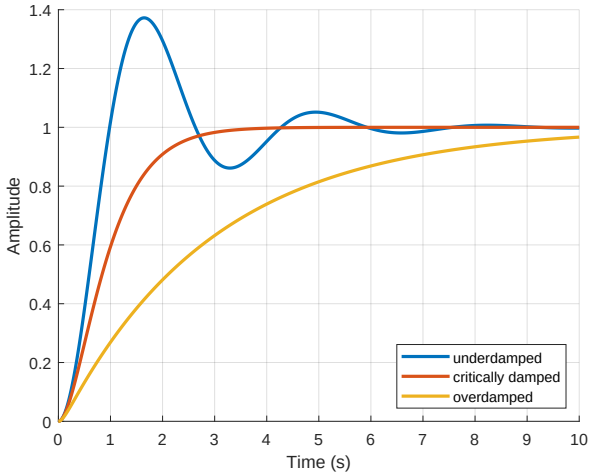


Figure 3.2: Step response of the PI scheme with relocated proportional action, for different value of the damping term.

only for the relocated scheme, because it has no zeros in its transfer function.

Conversely, first PI scheme we analyzed has a zero. Its response can be seen in Fig. 3.3, and it presents overshoot for $\zeta = 1$.

This behavior can be interpreted as the presence of a zero *speeding up* the response. In fact, if all parameters are kept equal, the relocated scheme shows significantly slower convergence speed.

3.3 Transient performance indicators

For the PI scheme with relocated proportional action, we can derive some performance indicators related to the transient in response to a step input.

Peak time This is the time necessary to reach the first peak of the response, i.e., the overshoot peak. We can find this value by deriving the expression of $\omega(t)$ and setting the derivative equal to zero.

$$\frac{d\omega}{dt} = \bar{\omega}_{\text{ref}} \left(-\sigma \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}} \sin(\omega_0 t + \phi) + \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}} \omega_0 \cos(\omega_0 t + \phi) \right) = 0 \quad (3.19)$$

$$\omega_0 \cos(\omega_0 t + \phi) - \sigma \sin(\omega_0 t + \phi) = 0 \quad (3.20)$$

Now recall that $\sigma = \zeta\omega_n$ and $\omega_0 = \omega_n\sqrt{1-\zeta^2}$

$$\sqrt{1-\zeta^2} \cos(\omega_0 t + \phi) - \zeta \sin(\omega_0 t + \phi) = 0 \quad (3.21)$$

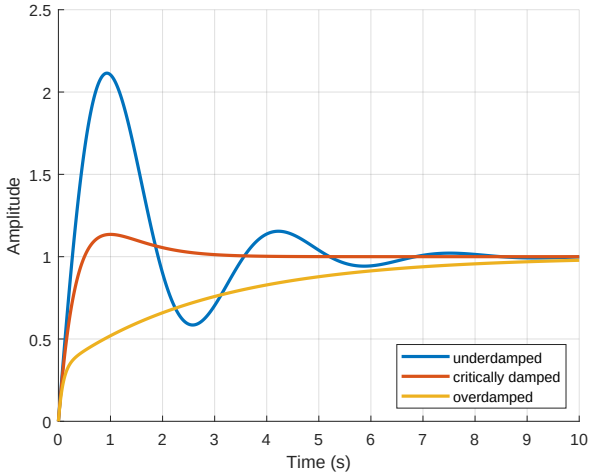


Figure 3.3: Step response of the standard PI scheme, for different value of the damping term. Since its transfer function has a zero, the critically damped response presents overshoot.

Recall that we picked ϕ such that $\cos\phi = \zeta$ and $\sin\phi = \sqrt{1-\zeta^2}$

$$\sin\phi\cos(\omega_0 t + \phi) - \cos\phi\sin(\omega_0 t + \phi) = 0 \tag{3.22}$$

Now recall that $\sin\alpha\cos\beta - \cos\alpha\sin\beta = \sin(\alpha - \beta)$, thus

$$\sin(\omega_0 t + \phi - \phi) = \sin(\omega_0 t) = 0 \tag{3.23}$$

This equation has infinite solutions, for $\omega_0 t = n\pi$. However, we are only interested in the first peak, so we can set $n = 1$ and find $t_p = \pi/\omega_0$, which means

$$\boxed{t_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}}} \tag{3.24}$$

Percent overshoot This value indicates the value of the first overshoot peak, expressed as a percentage of the reference value. To find this value we must substitute the peak time we just found inside the expression of $\omega(t)$.

$$\omega(t) = \bar{\omega}_{\text{ref}}\left(1 - \frac{e^{-\sigma\frac{\pi}{\omega_0}}}{\sqrt{1-\zeta^2}}\sin(\pi + \phi)\right) \tag{3.25}$$

We can simplify this expression by noting that $\sin(\pi + \phi) = -\sin\phi$, and that $\sin\phi$ simplifies with the $\sqrt{1-\zeta^2}$ in the denominator. In the exponent, we substitute $\sigma/\omega_0 = \zeta/\sqrt{1-\zeta^2}$.

$$\omega(t) = \bar{\omega}_{\text{ref}}\left(1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}\right) \tag{3.26}$$

The percent overshoot is the error over the reference, multiplied by 100%, i.e.,

$$\boxed{\text{OS}\% = \frac{|\bar{\omega}_{\text{ref}} - \omega|}{\bar{\omega}_{\text{ref}}} \cdot 100\% = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \cdot 100\%} \tag{3.27}$$

Settling time This is defined as the time necessary to reach 98% of the steady-state value. To find this value we have to make some approximations. At any given time t, the relative distance between the output $\omega(t)$ and the reference ω_{ref} is given by

$$\frac{|\omega_{\text{ref}} - \omega(t)|}{\omega_{\text{ref}}} = \frac{e^{-\sigma t}}{\sqrt{1-\zeta^2}} |\sin(\omega_0 t + \phi)|. \tag{3.28}$$

We impose this value to be around 2% of the reference value

$$\frac{e^{-\sigma t_s}}{\sqrt{1-\zeta^2}} |\sin(\omega_0 t + \phi)| = 0.02. \tag{3.29}$$

Since (3.29) doesn't have a close form solution we take the upperbound $|\sin(\omega_0 t + \phi)| \leq 1$. Physically, this means ignoring the fast oscillations and looking only at the pure exponential decay. We want to know the time t for which this value is at 0.02, therefore we impose

$$\frac{e^{-\sigma t_s}}{\sqrt{1-\zeta^2}} = 0.02. \tag{3.30}$$

After some manipulation

$$\begin{aligned} e^{-\sigma t_s} &= 0.02\sqrt{1-\zeta^2} \\ \Leftrightarrow -\sigma t_s &= \ln(0.02\sqrt{1-\zeta^2}) \\ \Leftrightarrow t_s &= -\frac{\ln(0.02\sqrt{1-\zeta^2})}{\sigma} \\ \Leftrightarrow t_s &= -\frac{\ln(0.02\sqrt{1-\zeta^2})}{\zeta\omega_n} \end{aligned} \tag{3.31}$$

When ζ is not close to $1 \Rightarrow \sqrt{1=\zeta^2} \approx 1$ and we have that

$$\boxed{t_s \approx -\frac{\ln(0.02)}{\zeta\omega_n}} \tag{3.32}$$

Physically, this means to have a very little damping, and a very fast convergence (together with wider oscillations), and we consider this as a lower bound for t_s.

3.4 Transient performance of the first order system

Transient performance is usually analyzed for the second order system, because of the possible presence of overshoot. However we could analyze it also for the first order system.

The first two indicators are trivial to analyze, because there is no overshoot. We could say that the peak time is $+\infty$ and that the percent overshoot is 0%.

The settling time however is meaningful, and can be analyzed by setting the output equal to 98% of the reference.

$$0.98 \cdot \bar{\omega}_{\text{ref}} = \bar{\omega}_{\text{ref}} \left(1 - e^{-\frac{k_p}{J} t_s}\right) \tag{3.33}$$

$$e^{-\frac{k_p}{J} t_s} = 0.02 \tag{3.34}$$

$$\boxed{t_s = -\frac{J}{k_p} \ln(0.02)} \tag{3.35}$$

This expression has the exact same form both for the first and the second order system, which is one of the reasons why we performed those approximation in the second order case.

Chapter 4

Position control

4.1 Motor model for position control

In position control, the goal is to achieve a target angular position for the rotating part of the motor. To derive a model useful for this purpose, let's write the angular speed as a derivative of the motor angular position $\omega = d\theta/dt$

$$J\frac{d^2\theta}{dt^2} = \tau_e - \tau_l - B\frac{d\theta}{dt} \tag{4.1}$$

and the Laplace transform:

$$s^2 J\Theta = T_e - T_l - sB\Theta \tag{4.2}$$

so the transfer function is

$$\frac{\Theta}{T_e - T_l} = \frac{1}{s(Js + B)} \tag{4.3}$$

which can also be expressed as the transfer function we used in the speed control scheme, in series with an integrator. This is of course to be expected, as the angular position is simply the integral of the motor speed.

4.2 Proportional control

Consider now a controller in which the control action is simply proportional to the error in the angular position. This of course is equivalent to the proportional speed control scheme, except that now the $W_m(s)$ is given by (4.3). The input-

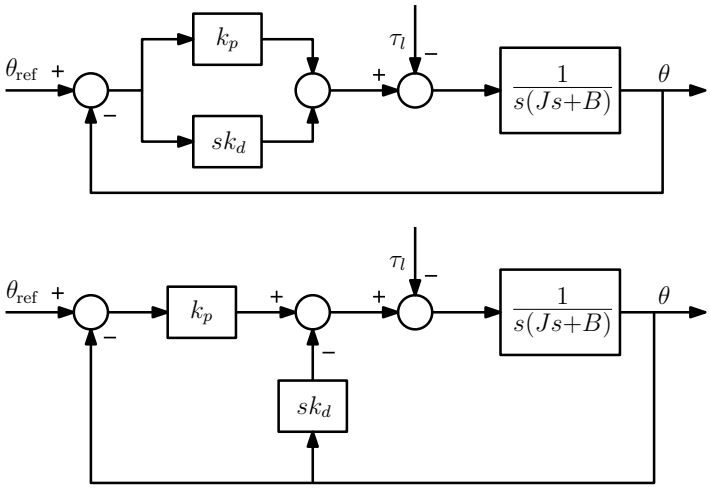


Figure 4.1: PD control scheme for position control (top), and relocated PD (bottom).

output transfer function of the closed-loop scheme is

$$\begin{aligned} W_{\text{i/o}}(s) &= \frac{W_c(s)W_m(s)}{1 + W_c(s)W_m(s)} \\ &= \frac{k_p/(s(Js+B))}{1 + k_p/(s(Js+B))} \\ &= \frac{k_p}{Js^2 + Bs + k_p} \end{aligned} \tag{4.4}$$

Using the final value theorem we can compute the steady-state value in response to a step input $\bar{\theta}_{\text{ref}} h(t)$

$$\begin{aligned} \lim_{t\to\infty} \theta(t) &= \lim_{s\to 0} s \cdot W(s) \cdot \frac{\bar{\theta}_{\text{ref}}}{s} \\ &= \lim_{s\to 0} s \frac{k_p}{Js^2 + Bs + k_p} \frac{\bar{\theta}_{\text{ref}}}{s} = \bar{\theta}_{\text{ref}} \end{aligned} \tag{4.5}$$

indicating that the target value will be reached regardless of the presence of friction. However, here we are ignoring the presence of a load torque. If we consider its effect, we must first compute the disturbance-output transfer function:

$$\begin{aligned} W_{\text{d/o}}(s) &= -\frac{W_m(s)}{1 + W_c(s)W_m(s)} \\ &= -\frac{1/(s(Js+B))}{1 + k_p/(s(Js+B))} \\ &= -\frac{1}{Js^2 + Bs + k_p} \end{aligned} \tag{4.6}$$

The steady-state value is then modified

$$\begin{aligned}\lim_{t\to\infty}\theta(t) &= \lim_{s\to 0} s\left(W_{\mathrm{i/o}}(s)\cdot\frac{\bar{\theta}_{\mathrm{ref}}}{s}+W_{\mathrm{d/o}}(s)\cdot\frac{\tau_l}{s}\right)\\ &= \lim_{s\to 0} s\left(\frac{k_p}{Js^2+Bs+k_p}\frac{\bar{\theta}_{\mathrm{ref}}}{s}-\frac{1}{Js^2+Bs+k_p}\frac{\tau_l}{s}\right)\\ &= \bar{\theta}_{\mathrm{ref}}-\frac{\tau_l}{k_p}\end{aligned} \tag{4.7}$$

There is something important to notice here. Assume that no friction is present: then the input/output transfer function takes the form

$$W_{\mathrm{i/o}}(s)=\frac{k_p}{Js^2+k_p} \tag{4.8}$$

Trying to apply the final value theorem to this expression would lead us to the wrong conclusion: in fact, the final value theorem requires that both $\lim_{t\to\infty} f(t)$ and $\lim_{t\to s} sF(s)$ exist. But since the above expression is the transfer function of an *undamped* oscillator, the limit $\lim_{t\to\infty} f(t)$ does **not** exist. Therefore, using the final value theorem would give us the wrong answer here. The step response of this system is actually an oscillating function.

Inapplicability of the Final Value Theorem for $B = 0$

Setting $B = 0$ in Eq. (4.8) and defining the natural frequency $\omega_n = \sqrt{k_p/J} > 0$, the input-output transfer function reduces to:

$$W_{i/o}(s)=\frac{k_p}{Js^2+k_p}=\frac{\omega_n^2}{s^2+\omega_n^2} \tag{4.9}$$

For a step reference $\Theta_{\mathrm{ref}}(s)=\frac{\bar{\theta}_{\mathrm{ref}}}{s}$, the Laplace transform of the output is $\Theta(s)=W_{i/o}(s)\Theta_{\mathrm{ref}}(s)$. Multiplying by s yields:

$$s\Theta(s)=\frac{\omega_n^2}{s^2+\omega_n^2}\bar{\theta}_{\mathrm{ref}} \tag{4.10}$$

To verify the true time-domain behavior, we expand $\Theta(s)$ into partial fractions:

$$\Theta(s)=\bar{\theta}_{\mathrm{ref}}\frac{\omega_n^2}{s(s^2+\omega_n^2)}=\bar{\theta}_{\mathrm{ref}}\left(\frac{A}{s}+\frac{Bs+C}{s^2+\omega_n^2}\right) \tag{4.11}$$

Equating numerators gives $\omega_n^2=(A_1+A_2)s^2+A_3s+A_1\omega_n^2$, which yields $A_1=1$, $A_2=-1$, and $A_3=0$:

$$\Theta(s)=\bar{\theta}_{\mathrm{ref}}\left(\frac{1}{s}-\frac{s}{s^2+\omega_n^2}\right) \tag{4.12}$$

Applying the inverse Laplace transform yields the time-domain step response for $t\geq 0$:

$$\theta(t)=\bar{\theta}_{\mathrm{ref}}\left(1-\cos(\omega_n t)\right) \tag{4.13}$$

As $t \to \infty$, $\cos(\omega_n t)$ oscillates perpetually in $[-1, 1]$, causing $\theta(t)$ to oscillate undamped in $[0, 2\bar{\theta}_{\text{ref}}]$. Therefore, $\lim_{t\to\infty} \theta(t)$ does not exist, and the limit $\lim_{s\to 0} s\Theta(s) = \bar{\theta}_{\text{ref}}$ merely represents the mean value of the oscillation.

4.3 Proportional + Derivative (PD) Action

In order to solve this issue, we can add a *derivative* action to the controller

$$\tau_e = k_p(\theta_{\text{ref}} - \theta) + k_d \frac{d(\theta_{\text{ref}} - \theta)}{dt} \tag{4.14}$$

The transfer function of this controller is

$$T_e = (k_p + k_d s)(\Theta_{\text{ref}} - \Theta) \tag{4.15}$$

The input/output transfer function with this controller is

$$\begin{aligned} W_{\text{i/o}}(s) &= \frac{W_c(s)W_m(s)}{1 + W_c(s)W_m(s)} \\ &= \frac{(k_p + k_d s)/(s(Js + B))}{1 + (k_p + k_d s)/(s(Js + B))} \\ &= \frac{k_p + k_d s}{Js^2 + Bs + k_p + k_d s} \\ &= \frac{k_p + k_d s}{Js^2 + (B + k_d)s + k_p} \end{aligned} \tag{4.16}$$

We can see that, even if we were to set $B = 0$, we would still get a damped response. One might argue that in a realistic setting friction is always present, and it is never really an option to set it equal to zero. This might be true, but it is certainly not a good choice to let the friction take care of the damping all by itself. Indeed, we want to be able to modulate the amount of damping in the response, so the derivative term will have to be used.

Similarly to the case of the PI, also here we have the option of using a relocated scheme, in which the derivative term is moved from the error to the feedback pathway. In this case, the output (without load) is

$$\Omega = W_m(k_p(\Theta_{\text{ref}} - \Theta) - s k_d \Theta) \tag{4.17}$$

$$\begin{aligned} W_{\text{i/o}}(s) &= \frac{k_p W_m(s)}{1 + (k_p + k_d s))W_m(s)} \\ &= \frac{1}{s(Js + B)} \left(\frac{k_p}{1 + (k_p + k_d s)/(s(Js + B))} \right) \\ &= \frac{k_p}{Js^2 + Bs + k_p + k_d s} \\ &= \frac{k_p}{Js^2 + (B + k_d)s + k_p} \end{aligned} \tag{4.18}$$

Similarly to the relocated PI, we get a second order response with no zeros. Therefore, its time response will be the same, bearing in mind that in this case the value of ω_n and ζ should be defined accordingly.

$$\omega_n = \sqrt{\frac{k_p}{J}}, \quad \zeta = \frac{B + k_d}{2J\omega_n} \tag{4.19}$$

Of course, since it does not have an integral action, this scheme would not be able to reject a constant load torque.

4.4 PI Action

Considering the relocated control scheme for PI control, the control action is

$$\Theta = W_m(\frac{k_i}{s}(\Theta_{\text{ref}} - \Theta) - k_p\Theta) \tag{4.20}$$

for which the input-output transfer function is

$$\begin{aligned} W_{\text{i/o}}(s) &= \frac{W_m(s)k_i/s}{1 + (k_p + k_i/s))W_m(s)} \\ &= \frac{1}{s^2(Js + B)} \left(\frac{k_i}{1 + ((k_p s + k_i)/s)/(s(Js + B))} \right) \\ &= \frac{k_i}{s^2(Js + B) + k_p s + k_i} \\ &= \frac{k_i}{Js^3 + Bs^2 + k_p s + k_i} \end{aligned} \tag{4.21}$$

that can used to reject a constant load but suffer from the same problem as before if $B = 0$.

4.5 PID controller

We solve both problems by not using PD or PI schemes alone, but combining them. The resulting control action is a PID that has the following transfer function with no relocation:

$$\begin{aligned} W_{\text{i/o}}(s) &= \frac{W_c(s)W_m(s)}{1 + W_c(s)W_m(s)} \\ &= \frac{(k_p + k_d s + k_i/s)/(s(Js + B))}{1 + (k_p + k_d s + k_i/s)/(s(Js + B))} \\ &= \frac{k_p + k_d s + k_i/s}{Js^2 + Bs + k_p + k_d s + k_i/s} \\ &= \frac{k_d s^2 + k_p s + k_i}{Js^3 + (B + k_d)s^2 + k_p s + k_i} \end{aligned} \tag{4.22}$$

Chapter 5

Feedforward compensation

5.1 Feedforward speed control

Consider the motor model without load torque:

$$J\frac{d\omega(t)}{dt} = T_e(t) - B\omega(t). \tag{5.1}$$

Assume that we know an approximate model of the motor

$$\hat{J}\frac{d\omega(t)}{dt} = T_e(t) - \hat{B}\omega(t) \tag{5.2}$$

where $\hat{J}$ and $\hat{B}$ are approximate parameters for the motor inertia and friction coefficient.

If there is no load torque, the electromotive torque to be applied in order to realize a given reference can be approximately computed as

$$T_{\text{ff}}(t) = \hat{J}\frac{d\omega_{\text{ref}}(t)}{dt} + \hat{B}\omega_{\text{ref}}(t) \tag{5.3}$$

Let's look at the transfer function of the feedforward action. Taking the transform of both sides we get

$$T_{\text{ff}}(s) = (\hat{J}s + \hat{B})\Omega_{\text{ref}}(t) \tag{5.4}$$

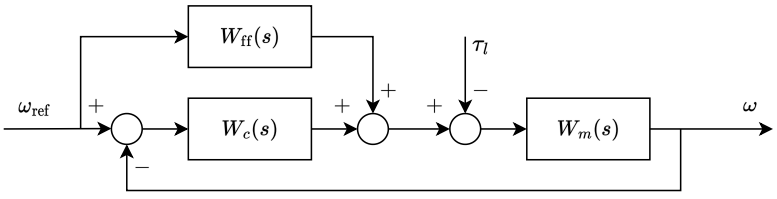


Figure 5.1: A scheme with feedforward action.

so the transfer function is

$$W_{\text{ff}}(s) = \frac{T_{\text{ff}}(s)}{\Omega_{\text{ref}}(t)} = \hat{J}s + \hat{B} \tag{5.5}$$

Let's see at the resulting closed-loop transfer function with the feedforward action, as in Fig. (5.1), scheme (a). The output signal is

$$\begin{aligned} \Omega(s) &= W_m(s)(T_e(s) + T_{\text{ff}}(s)) \\ &= W_c(s)W_m(s)(\Omega_{\text{ref}}(s) - \Omega(s)) + W_{\text{ff}}(s)W_m(s)\Omega_{\text{ref}}(s) \end{aligned} \tag{5.6}$$

Group terms:

$$(1 + W_c(s)W_m(s))\Omega(s) = (W_c(s)W_m(s) + W_{\text{ff}}(s)W_m(s))\Omega_{\text{ref}}(s) \tag{5.7}$$

and compute the transfer function:

$$W_{\text{i/o}}(s) = \frac{\Omega(s)}{\Omega_{\text{ref}}(s)} = \frac{W_c(s)W_m(s) + W_{\text{ff}}(s)W_m(s)}{1 + W_c(s)W_m(s)} \tag{5.8}$$

from which we can see that if we pick $W_{\text{ff}}(s) = W_m^{-1}(s)$, we get $W_{\text{i/o}}(s) = 1$. In fact, if $\hat{B} = B$ and $\hat{J} = J$, the feedforward transfer function is the inverse of the motor transfer function.

If the load torque τ_l is present and known, its estimated value $\hat{\tau}_l$ can be added to the computation of the feedforward term:

$$\tau_{\text{ff}}(t) = \hat{J}\frac{d\omega_{\text{ref}}(t)}{dt} + \hat{B}\omega_{\text{ref}}(t) + \hat{\tau}_l(t) \tag{5.9}$$

Compute the Laplace transform of the feedforward action

$$T_{\text{ff}}(s) = \hat{J}s\Omega_{\text{ref}}(s) + \hat{B}\Omega_{\text{ref}}(s) + \hat{T}_l(s) \tag{5.10}$$

The output signal is now

$$\begin{aligned} \Omega(s) &= W_m(s)(T_e(s) + T_{\text{ff}}(s) - T_l(s)) \\ &= W_c W_m(\Omega_{\text{ref}} - \Omega) + W_{\text{ff}} W_m \Omega_{\text{ref}} + W_m(\hat{T}_l - T_l) \end{aligned} \tag{5.11}$$

Group terms:

$$(1 + W_c W_m)\Omega = (W_c W_m(s) + W_{\text{ff}} W_m)\Omega_{\text{ref}} + W_m(\hat{T}_l - T_l) \tag{5.12}$$

from which we find the output signal in closed loop.

$$\Omega(s) = \frac{W_c W_m + W_{\text{ff}} W_m}{1 + W_c W_m}\Omega_{\text{ref}} + \frac{W_m}{1 + W_c W_m}(\hat{T}_l - T_l) \tag{5.13}$$

still, if we pick $W_{\text{ff}} = W_m^{-1}(s)$, the input-output transfer function is 1, which corresponds to the first term. In addition, if our knowledge of the load disturbance is perfect, we achieve perfect disturbance rejection, which means that Ω is perfectly replicated (as long as $\omega(0) = \omega_{\text{ref}}(0)$).

5.2 Initial condition not matched

Let's now see what happens if the initial condition of the reference we want to track does not match the initial condition of our system. We will in particular assume, as always that the motor starts at $\omega(0) = 0$, but that the reference is a $\omega_{\text{ref}}(t)$ such that $\omega_{\text{ref}}(0) \neq 0$.

The difference with respect to the previous case is in the fact that when we compute the feedforward term we have to use the correct expression when computing the derivative of ω_{ref}. Recall that the Laplace transform of the derivative of a function is actually

$$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0) \tag{5.14}$$

Therefore the transform of the feedforward action is

$$T_{\text{ff}}(s) = \hat{J}(s\Omega_{\text{ref}}(s) - \omega_{\text{ref}}(0)) + \hat{B}\Omega_{\text{ref}}(s) + \hat{T}_l(s) \tag{5.15}$$

where we can name as W_{ff} the transfer function of the feedforward term, and the remaining term we leave as additive contributions

$$\begin{aligned} T_{\text{ff}}(s) &= (\hat{J}s + \hat{B})\Omega_{\text{ref}}(s) - \hat{J}\omega_{\text{ref}}(0) + \hat{T}_l(s) \\ &= W_{\text{ff}}(s)\Omega_{\text{ref}}(s) - \hat{J}\omega_{\text{ref}}(0) + \hat{T}_l(s) \end{aligned} \tag{5.16}$$

The output signal can now be written as

$$\begin{aligned} \Omega(s) &= W_m(s)(T_e(s) + T_{\text{ff}}(s) - T_l(s) - \hat{J}\omega_{\text{ref}}(0)) \\ &= W_c W_m(\Omega_{\text{ref}} - \Omega) + W_{\text{ff}} W_m \Omega_{\text{ref}} + W_m(\hat{T}_l - T_l - \hat{J}\omega_{\text{ref}}(0)) \end{aligned} \tag{5.17}$$

Group terms:

$$(1 + W_c W_m)\Omega = (W_c W_m(s) + W_{\text{ff}} W_m)\Omega_{\text{ref}} + W_m(\hat{T}_l - T_l - \hat{J}\omega_{\text{ref}}(0)) \tag{5.18}$$

from which we find the output signal in closed loop.

$$\Omega(s) = \frac{W_c W_m + W_{\text{ff}} W_m}{1 + W_c W_m}\Omega_{\text{ref}} + \frac{W_m}{1 + W_c W_m}(\hat{T}_l - T_l - \hat{J}\omega_{\text{ref}}(0)) \tag{5.19}$$

Now assume perfect system knowledge ($\hat{J} = J$, $\hat{B} = B$) and perfect load knowledge ($\hat{T}_l = T_l$). In this case, we have $W_{\text{ff}} = W_m^{-1}$, so the output is

$$\Omega(s) = \Omega_{\text{ref}} + \frac{W_m}{1 + W_c W_m} J\omega_{\text{ref}}(0) \tag{5.20}$$

from which we can compute the error

$$E(s) = \Omega_{\text{ref}} - \Omega(s) = -\frac{W_m}{1 + W_c W_m} J\omega_{\text{ref}}(0) \tag{5.21}$$

Let's for example assume that we are using a proportional controller, thus $W_c = k_p$

$$\begin{aligned} E(s) &= -\frac{W_m}{1 + W_c W_m} J\omega_{\text{ref}}(0) \\ &= -\frac{1/(Js + B)}{1 + k_p/(Js + B)} J\omega_{\text{ref}}(0) \\ &= -\frac{1}{s + (B + k_p)/J}\omega_{\text{ref}}(0) \end{aligned} \tag{5.22}$$

This is just the Laplace transform of a decreasing exponential term, i.e., its time expression is

$$e(s) = \omega_{\text{ref}}(0)e^{-t/\tau} \tag{5.23}$$

where $\tau = J/(B + k_p)$. Therefore, the initial error due to the wrong initial condition is completely recovered, and the controller is able to perfectly track the reference at steady state. We can also see this by using the final value theorem:

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} \left(-\frac{1}{s + (B + k_p)/J}\omega_{\text{ref}}(0) \right) s = 0 \tag{5.24}$$

5.3 Feedforward position control

If the goal is to achieve position control, we can also use feedforward. This time, we need to use the model expressed in terms of the angle, which has both the derivative and the second derivative of the output.

$$\tau_{\text{ff}}(t) = \hat{J}\frac{d^2\theta_{\text{ref}}}{dt^2} + \hat{B}\frac{d\theta_{\text{ref}}}{dt} + \hat{\tau}_l \tag{5.25}$$

which in the frequency domain becomes

$$T_{\text{ff}}(s) = \hat{J}s^2\Theta_{\text{ref}} + \hat{B}s\Theta_{\text{ref}} + \hat{T}_l \tag{5.26}$$

and therefore can be expressed as

$$T_{\text{ff}}(s) = (\hat{J}s^2 + \hat{B}s)\Theta_{\text{ref}} + \hat{T}_l = W_{ff}\Theta_{\text{ref}} + \hat{T}_l \tag{5.27}$$

where $W_{ff}(s) = \hat{J}s^2 + \hat{B}s$ is the transfer function of a feedforward block, and it is the inverse of the model, i.e., $W_{\text{ff}} = W_m^{-1}$.

This too will give us a closed-loop transfer function equal to 1, if the model is perfectly known. However, note that if there are imprecisions, such as for example a non-matched initial condition, this will generate an error response that is different from the one we saw for the feedforward for speed control case. In particular, using a proportional control only will result in an undamped (or only damped by friction) system, and thus require adding a derivative term, just as we saw in the position control case.

5.4 Partial feedforward

Sometimes we are unable to achieve complete compensation of every term in the dynamic equation. However we may still be able to compensate parts of it. For example, we may not have good estimates for the model parameters $\hat{J}$ and $\hat{B}$, but we may have a good estimate of the load torque $\hat{\tau}_l$, and therefore we may choose to only compensate the latter. In this case, the feedforward term would only consist of

$$\tau_{\text{ff}}(t) = \hat{\tau}_l(t) \tag{5.28}$$

Another example is one case is one in which we are performing set-point regulation for speed control. In this case we **cannot** use feedforward, because the reference is a step function and therefore it can't be differentiated in $t = 0$. However, we can still compensate the friction term:

$$\tau_{\text{ff}}(t) = \hat{B}\theta_{\text{ref}}(t) \tag{5.29}$$

This will not give us perfect tracking of the reference (because perfect tracking of a step reference cannot be achieved anyway), but might improve the steady-state behavior. However, note that this particular approach to friction compensation is quite simplistic, and it is only given here as it follows nicely as an application of feedforward, but actual friction compensation might be more complex, and it is not part of this course.

Chapter 6

Discrete control

6.1 Sampling

Sampling is a fundamental concept in signal processing that enables us to convert continuous-time signals into discrete-time representations. The Shannon-Nyquist Theorem guides how we sample continuous signals accurately and avoid issues such as aliasing.

6.1.1 Shannon-Nyquist Theorem

To accurately reconstruct a continuous signal from its samples, the sampling rate must be at least twice the highest frequency component of the signal.

Mathematically, this can be expressed as

$$f_s \geq 2f_{\max} \tag{6.1}$$

where f_s is the sampling frequency (in Hz) and $f_{\max}$ The maximum frequency component in the continuous signal (in Hz).

6.1.2 Aliasing

Aliasing occurs when a signal contains frequency components above the Nyquist frequency, which is half the sampling frequency ($f_s/2$), and is insufficiently sampled (see Fig. 6.1) In this scenario, the original higher-frequency components "fold" or alias into the lower-frequency range, leading to an incorrect representation of the signal. This folding of frequencies can result in severe distortion and misinterpretation of the signal.

The graphical representation of aliasing is often seen as an overlapping or superimposition of the folded frequencies onto the lower-frequency components. This overlap can make it challenging to distinguish between the original signal and the aliased components.

Tools to prevent aliasing:

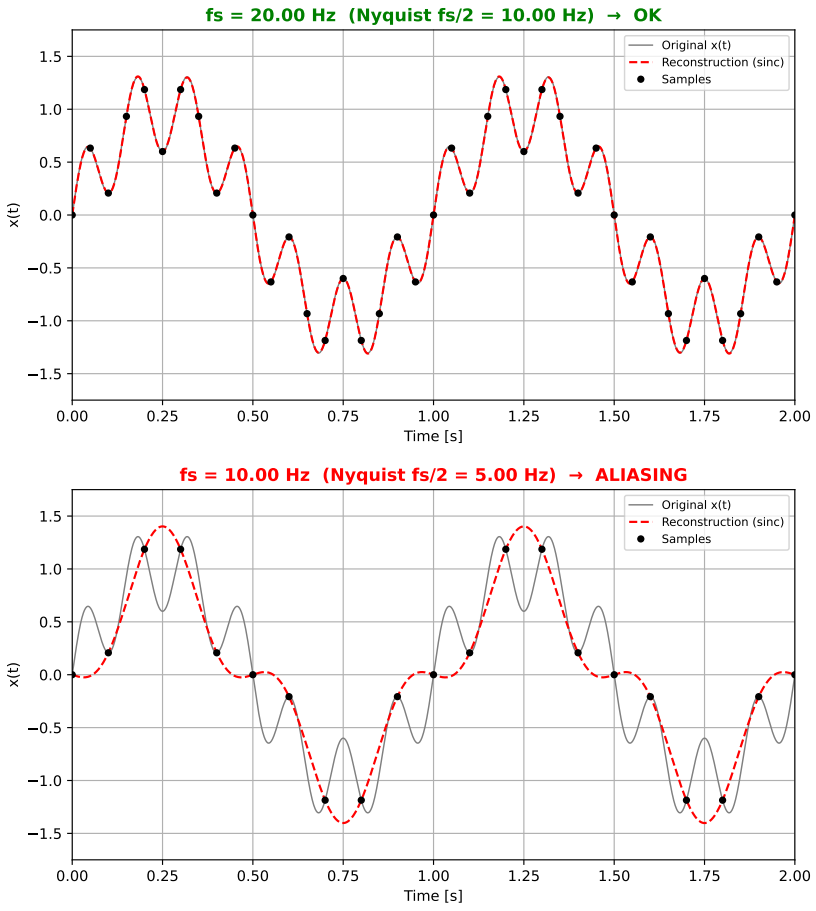


Figure 6.1: Aliasing effect by the sampling of a continuous signal $x(t) = \sum A_i \sin(2\pi f_i t + \varphi_i)$ with a frequency $f_i \in [1.0, 7.0]$.

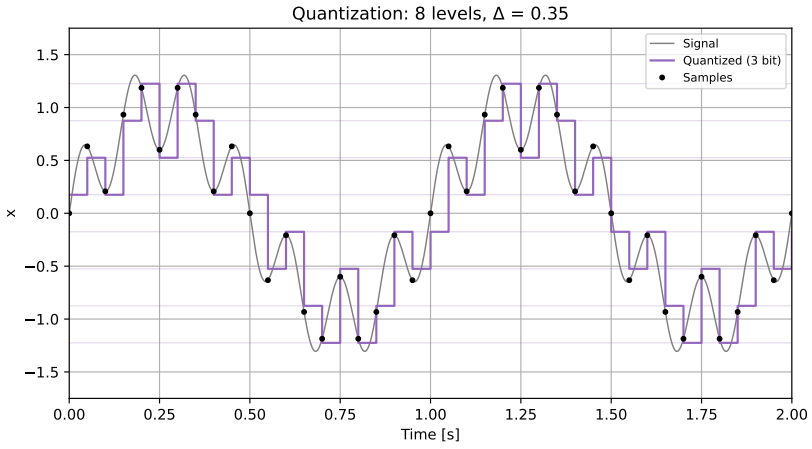


Figure 6.2: Quantization error of the same signal with a sampling freq of $f_s = 20$

- Band-Limited Signals: Ensure that the original signal is band-limited, meaning it does not contain frequency components beyond the Nyquist frequency.
- Anti-Aliasing Filters: Prior to sampling, apply anti-aliasing filters to attenuate or remove frequencies above the Nyquist frequency. These filters ensure that only the relevant frequency components are captured during sampling.
- Higher Sampling Rates: Increase the sampling rate if the signal contains high-frequency components. A higher sampling rate provides a more accurate representation of the signal.

6.1.3 Quantization and bit depth

When an analog signal is sampled and converted into a digital format, it undergoes a process called quantization (see Fig. 6.2). Quantization involves mapping each sampled value to a discrete digital value from a finite set of possible levels. The number of distinct levels or values that can be used to represent each sample is determined by the sampling resolution, typically measured in bits.

Sampling resolution defines precision of this quantization process. It determines how finely the analog signal is divided into discrete steps. Higher sampling resolution corresponds to smaller steps and finer precision, while lower resolution results in larger steps and coarser precision.

Sampling resolution is referred to as *bit depth*. For instance, an 8-bit resolution implies that there are 2^8 (256) possible digital values to represent each sample. Similarly, a 16-bit resolution provides 2^{16} (65536) possible values.

The difference between the original signal and the quantized one is the *sampling noise*, and the ratio between the signal and the noise level is the *signal-to-noise ratio*.

6.2 The z-transform

The way we model a sampled signal $(u_0, u_1, u_2, \dots)$ is by considering it as a sequence of Dirac delta, each shifted by an additional sample T, and each multiplied by its corresponding sampled value:

$$u(t) = u_0\delta(t) + u_1\delta(t-T) + u_2\delta(t-2T) + \dots \tag{6.2}$$

For this signal, we can compute its Laplace transform,

$$\mathcal{L}\{u(t)\} = U(s) = \sum_{k=0}^{\infty} u_k e^{-ksT} \tag{6.3}$$

In order to save up on notation, we can rename $z = e^{sT}$

$$U(z) = \sum_{k=0}^{\infty} u_k z^{-k} \tag{6.4}$$

This is the z-transform. It is simply the Laplace transform of a sampled signal, with an additional change of variable from s to z.

6.2.1 Pulse function

For example, if we consider a single pulse, such as

$$u(k) = \begin{cases} 1 & \text{if } k = 0 \\ 0 & \text{otherwise} \end{cases} \tag{6.5}$$

gives a discrete Laplace transform which is simply $U(z) = 1$.

6.2.2 Step function

A discrete step function

$$u(k) = \begin{cases} 1 & \text{if } k \geq 0 \\ 0 & \text{otherwise} \end{cases} \tag{6.6}$$

gives

$$U(z) = 1 + z^{-1} + z^{-2} + \dots = \sum_{i=0}^{\infty} z^{-i} = \frac{1}{1-z^{-1}} = \frac{z}{z-1} \tag{6.7}$$

6.2.3 Ramp function

$$\mathcal{L}[i] = 0 \cdot 1 + 1 \cdot z^{-1} + 2 \cdot z^{-2} + \cdots = \frac{z}{(z-1)^2} \tag{6.8}$$

6.2.4 Z-transform of delayed signal

Consider a signal $u(k)$. Its transform is defined as

$$U(z) = \sum_{i=0}^{\infty} u_i z^{-i} \tag{6.9}$$

Now shift the signal forward in time by one sample, which means that in the first position now we have 0, then we get u_1, then u_2, etc...

$$0 \cdot 1 + u_0 z^{-1} + u_1 z^{-2} + \cdots = \sum_{i=0}^{\infty} u_{i+1} z^{-i} = \mathcal{L}[u(k+1)] \tag{6.10}$$

6.2.5 Final value theorem

The final value theorem for the z-transform is as follows

$$\lim_{k \to \infty} f(k) = \lim_{z \to 1} \frac{z-1}{T} F(z) \tag{6.11}$$

6.3 Bilinear transformation

Since we perform all our analysis in the continuous domain s, we would like to have a way to convert a transfer function in s to a transfer function in z, and possibly conserve as much as possible the properties of the original system. We will consider three ways in which we can approximate the complex variable s, in order to obtain a discrete-time transfer function in the variable z.

6.3.1 Forward integration

$$\boxed{s = \frac{1}{T}(z-1)} \tag{6.12}$$

With this choice, the integrator would map to

$$\frac{1}{s} = T \frac{1}{z-1} \tag{6.13}$$

therefore, if we apply it as a transfer function we find

$$Y(z) = T \frac{1}{z-1} U(z) \implies Y(z)(z-1) = TU(z) \tag{6.14}$$

recall that when we go back to the time domain, $Y(z)$ is $y(k)$ and $zY(z)$ is $y(k+1)$. Therefore this transformation corresponds to interpreting the continuous time integrator $1/s$, as the discrete expression

$$y(k+1) - y(k) = T \cdot u(k) \implies y(k+1) = y(k) + T \cdot u(k) \tag{6.15}$$

which is effectively a forward integration with zero-order hold. We can rewrite in terms of the current step as

$$y(k) = y(k-1) + T \cdot u(k-1) \tag{6.16}$$

6.3.2 Backward integration

$$\boxed{s = \frac{1}{T}\frac{z-1}{z}} \tag{6.17}$$

In this case, the integrator is

$$\frac{1}{s} = T\frac{z}{z-1} \tag{6.18}$$

$$Y(z) = T\frac{z}{z-1}U(z) \implies Y(z)(z-1) = TzU(z) \tag{6.19}$$

$$y(k+1) = y(k) + T \cdot u(k+1) \tag{6.20}$$

We can rewrite in terms of the current step as

$$y(k) = y(k-1) + T \cdot u(k) \tag{6.21}$$

6.3.3 Tustin method

$$\boxed{s = \frac{2}{T}\frac{z-1}{z+1}} \tag{6.22}$$

In this case, the integrator is

$$\frac{1}{s} = \frac{T}{2}\frac{z+1}{z-1} \tag{6.23}$$

$$Y(z) = \frac{T}{2}\frac{z+1}{z-1}U(z) \implies Y(z)(z-1) = \frac{T}{2}(z+1)U(z) \tag{6.24}$$

$$y(k+1) = y(k) + T\frac{u(k) + u(k+1)}{2} \tag{6.25}$$

which means that it is using the average between the current and future input to compute the integral. This corresponds to trapezoidal integration. We can rewrite in terms of the current step as

$$y(k) = y(k-1) + T\frac{u(k+1) + u(k)}{2} \tag{6.26}$$

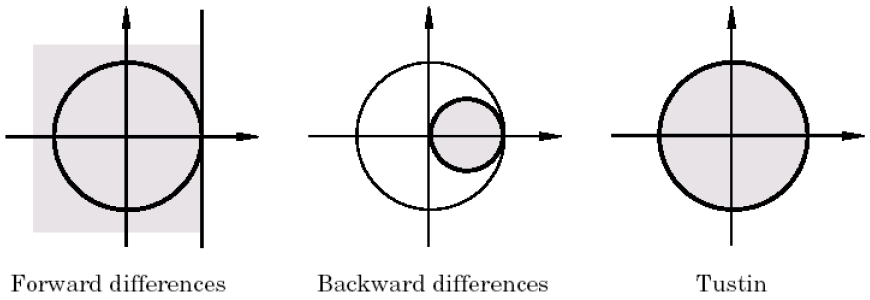


Figure 6.3: Mapping of stable poles from the continuous domain s to the discrete domain z.

6.3.4 Poles

In z, poles are stable if they are within the unit circle. One might ask whether discretizing with either method will preserve the stability of the system. The results are in fact different depending on which method is used, and are shown in Fig. 6.3. In particular, forward integration (left), maps stable poles in s to a half plane with real part less than 1. Thus, the resulting discrete pole might be stable, but might be not. In particular, it will be outside of the unit circle, if the sampling time T is large. Backward integration will map all poles inside of the unit circle, but will not cover the entire unit circle: rather all stable poles in s will be mapped to a circle of radius 0.5 with center in $(0.5, 0)$. Tustin's method, however, will exactly map all stable poles in s to the unit circle in z.

Chapter 7

Windup

7.1 Windup phenomenon

Windup is a nonlinear phenomenon that occurs when there is saturation in the actuation stage of the motor. Saturation will occur at some point because the actuation stage is typically designed to limit the torque that is being requested to the motor, with the goal of avoiding excessive values that might cause damage.

We will model torque saturation as an additional block in front of the motor model performing the following nonlinear transformation:

$$\text{sat}(\tau) = \begin{cases} \tau & \text{if } \tau \in [-\tau_{\max}, \tau_{\max}] \\ \tau_{\max} & \text{if } \tau > \tau_{\max} \\ -\tau_{\max} & \text{if } \tau < -\tau_{\max} \end{cases} \tag{7.1}$$

For the purpose of the following analysis, we will assume that the value $\tau_{\max}$ is known, as it is being either estimated or it is part of the specification of the motor.

To understand how windup works, consider a motor whose friction can be neglected. The model is therefore $W_m(s) = 1/(Js)$, which is an integrator.

When the actuation stage is saturated, the torque $\text{sat}(\tau)$ entering the motor is a constant value $\tau_{\max}$. Therefore, the speed profile will be the integral of this constant value, i.e., it will grow linearly as $J\tau_{\text{sat}} \cdot t$.

Let's consider that the controller is now a proportional-integral controller. At $t = 0$, the motor starts at zero speed and the reference is given as a step function. Therefore, if the reference value is large, the error is large and the actuator is likely to saturate. This will generate a linearly growing velocity profile, and the integral term will keep charging up until the motor arrives at the reference value, because until then the error is always positive.

Upon reaching the reference, the integral term will start decreasing because the error is negative. However, all of the accumulated integral is still a large positive value, therefore, the speed will keep growing linearly because the output of the controller is still a large positive value and it is being saturated. In order

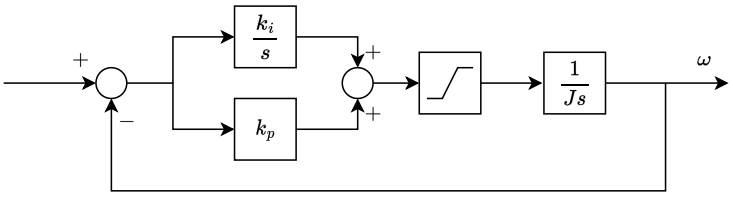


Figure 7.1: A PI scheme with a saturation stage will likely be subject to windup when a large reference is given.

for the speed profile to change direction and start heading towards the reference once again, we must wait for the integrator to discharge.

The result is a large increase in the overshoot of the system, compared to the case in which no saturation is present. This additional overshoot is highly undesirable, and it is only due to the fact that the integrator keeps integrating while the output is saturated, and the amount of charge it builds up is excessively large.

7.2 Anti windup

A simple anti-windup system can be built in the following way:

- First, we need to compare the output signal of the controller with the saturated signal that is entering the motor. To do this, we add a model of the saturation to our controller, and we perform a difference between τ_e and $\text{sat}(\tau_e)$
- This difference represents *how much* the output of the controller is being saturated, and therefore indicates how likely we are to observe windup
- Therefore, we can feedback this signal into the integral path of the controller, in order to reduce the amount of signal that is being integrated when saturation occurs. To do this, we multiply it by a gain and subtract it from the error signal before the integration.

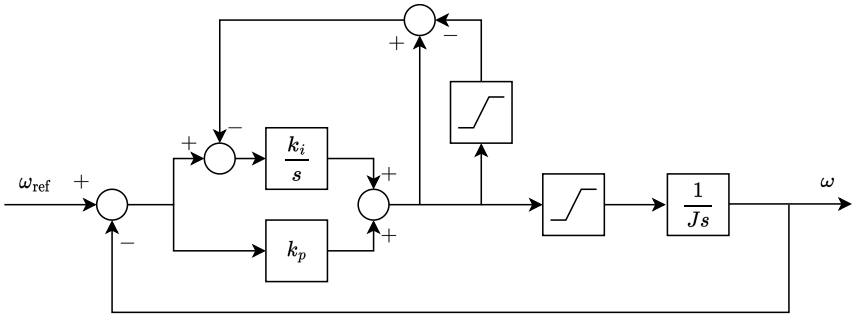


Figure 7.2: An anti-windup scheme.

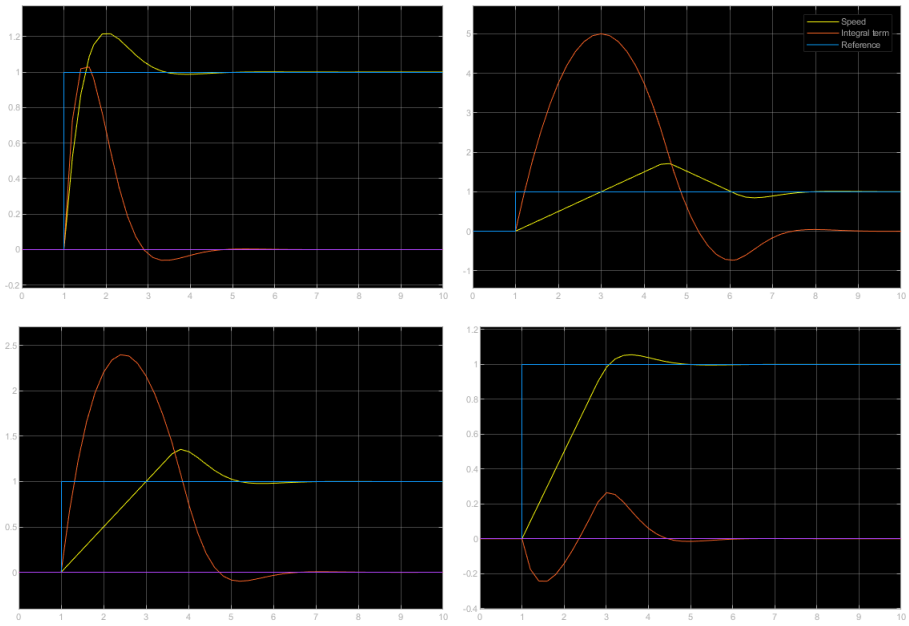


Figure 7.3: Windup example. In blue, a unit step reference; in yellow, the system response; in red, the value of the integral term over time. In the first plot, the response of the system without saturation. In the second plot, the response with saturation, which shows a large amount of windup, as the overshoot is almost as high as the reference. In the third plot we add an antiwindup system, and are able to reduce the amount of overshoot. In the fourth plot we increase the gain of the antiwindup and finally obtain a much better output transient.